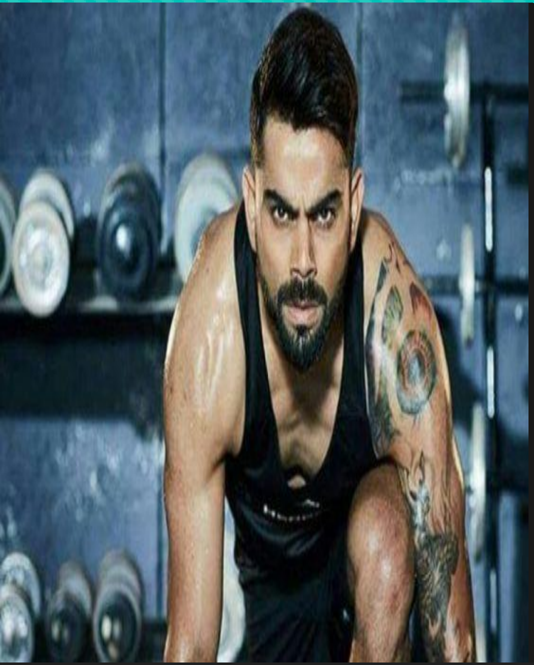
CAREER

MOST IMPORTANT THING IN YOUR LIFE ??

HEALTH

- Food
- Physical environment
- Social environment
- Personal hygiene
- Exercise

FOOD, NUTRITION, AND HYGIENE: BCHO 0011



Break Fast: omelette, Cheese, Spinach, Fruits

Lunch: Fresh Vegetables, Protein, Chicken, Potatoes,

Dinner: Fresh See food

Module-I

Concept Of Food and Nutrition

(a) Definition of Food, Nutrients, Nutrition, Health, balanced Diet
(b) Types of Nutrition- Optimum Nutrition, under Nutrition, Over Nutrition
(c) Meal planning- Concept and factors affecting Meal Planning
(d) Food groups and functions of food

Nutrients: Macro and Micro

RDA, Sources, Functions, Deficiency and excess of

(a) Carbohydrate
(b) Fats
(c) Proteins
(d) Minerals
Major: Calcium, Phosphorus, Sodium, Potassium
Trace: Iron, Iodine, Fluorine, Zinc
(e) Vitamins
Water Soluble : Vitamin B, C
Fat Soluble: Vitamin A, D, E, K
(f) Water
(g) Dietary Fiber

Module-II

1000 days Nutrition

(a) Concept, Requirement, Factors affecting growth of child

(b) Prenatal Nutrition (0 - 280 days): Additional Nutrients' Requirement and risk factors during pregnancy

(c) Breast / Formula Feeding (Birth – 6 months of age)

Complementary and Early Diet (6 months – 2 years of age)

Community Health Concept

(a) Causes of common diseases prevalent in the society and Nutrition requirement in the following:

Diabetes

Hypertension (High Blood Pressure)

Obesity

Constipation

Diarrhea Typhoid

(b) National and International Program and Policies for improving Dietary Nutrition

(c) Immunity Boosting Food

Book (1) Singh, Anita, "Food and Nutrition", Star Publication, Agra, India, 201

(2) https://pediatrics.aappublications.org/content/141/2/e201737168

(3) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5750909/.

(4)https://www.google.co.in/books/edition/Food_Nutrition_and_Hygiene_According_to/k_NIEAAAQBAJ?hl=en&gbpv=1

WHAT IS FOOD

?????

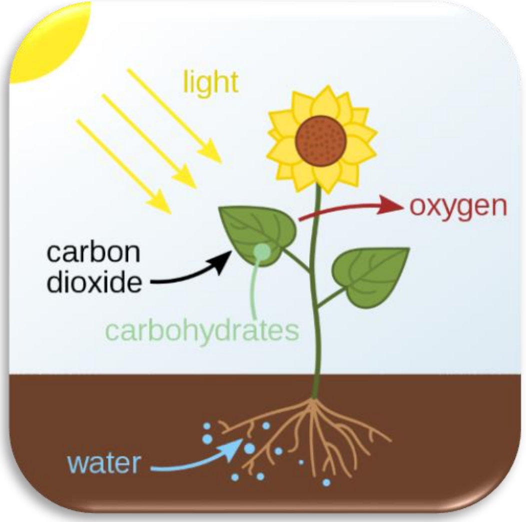
Food is the substance that we eat and satisfy hunger

What is Food ?????

Food is the substance that satisfies our (or organisms) hunger (or thirst) and provides nutrients after its digestion and assimilation in the body. These foods also supply energy, and enable the growth and repair of tissues and organs. It also protects the body from disease and regulates body functions.

Functions of food

- Satisfy Hunger
- Helps in Growth
- Repair tissues & Organ
- Protect from Disease
- Supply Energy
- Supply Nutrients



Photosynthesis

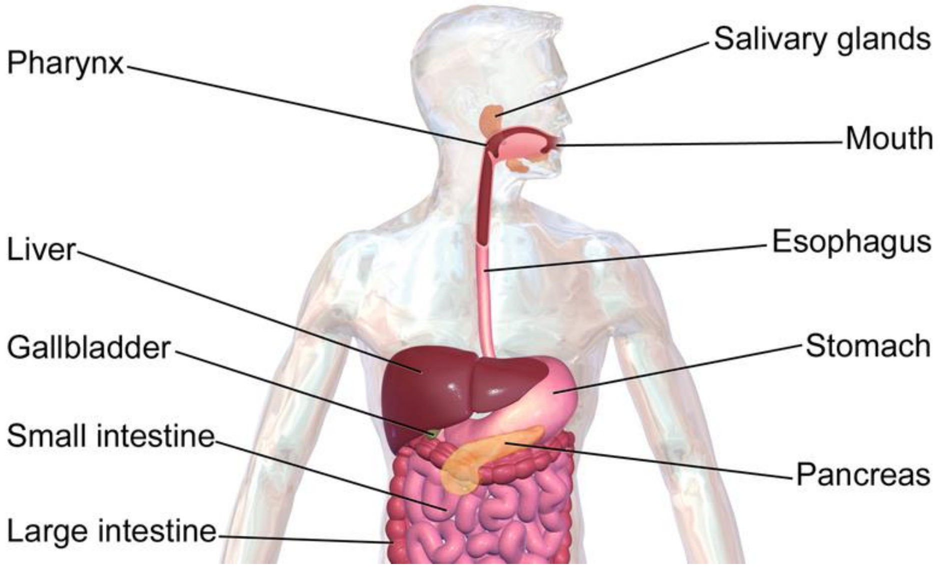
$$CO_2 + H_2O \xrightarrow[\text{Light}]{\text{Photosynthesis}} C_6H_{12}O_6 + 6O_2$$

Carbon dioxide + water → Glucose + Oxygen

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + ATP$$

Glucose + Oxygen → Carbon dioxide + water

Adenosine triphosphate (ATP) is the source of energy for use and storage at the cellular level (**Cellular energy currency**)



1. During chewing, food is broken down by the teeth inside the mouth

2. Those broken pieces of food mixed with saliva (It contains the digestive enzyme **amylase** which is capable of digesting **carbohydrates**)

3. This broken food transfer to the stomach through esophagus

4. In stomach HCl, **Pepsin**, **Lipase** are present in stomach. Food further broken down small in pieces (digestion)

5. Food is broken down into small pieces in the stomach and these small pieces are absorbed in the small intestine (this is called assimilation).

What is Nutrients

Essential chemical substances, generally obtained from food, which are essential for different activities(growth, immunity etc.) of the body and maintain health are called nutrients.

Food contains many complex chemical substances, which are necessary for the growth of the body, repair of tissue, protection from diseases and to regulate body function. Most food contains more than one nutrient e.g. food having protein, fat, etc.

The major nutrients in our food are carbohydrates, fats, proteins, vitamins, minerals and water.

What is Nutrition

According to D.F. Turner, “Nutrition is the combination of processes by which the living organism receives and utilizes the materials necessary for the maintenance of its functions, growth, and renewal of its component.”

Nutrition includes food intake, digestion, absorption, assimilation and excretion of waste.

In India, National Nutrition Week is celebrated each year from the 1^{st} to the 7^{th} of September. The purpose is to make people aware of important tips for their health and well-being.

What is Health

Health literally means, “that state of the body and mind in which it can perform all functions properly.” According to the World Health Organization (WHO), “health is the simple protection of the body from diseases”. Health is not merely the absence of disease and infirmity but refers to being healthy in a complete physical, mental and social state.

Although there are many types of health, these three types of health are among the ones approved by WHO:

Physical health-It means all the organs and cells of the body are functioning according to their expected capacity keeping in perfect harmony with the body.

Mental health-The correct harmony of man with the urban world i.e. the condition in which man can maintain harmony with himself and with the people and environment around him.

Social health-To places oneself in society recognize society as a larger form of one’s family and create such an environment for living.

Relation between food and Health

Foods are essential ingredients for our health. The contribution of food to our physical, mental, and social health are describe as bellow

Physical Health:

Nutrient Supply: Foods like fruits, vegetables, and lean proteins provide essential nutrients for growth, tissue repair, and overall body function.

Energy Source: Carbohydrates and fats in foods are the primary energy sources for daily activities.

Protect from diseases: protein, vitamins and minerals protect from different diseases. For example, vitamin A prevents night blindness and vitamin D prevents rickets diseases.

Mental Health:

Brain Function: Omega-3 fatty acids found in fish contribute to cognitive function and memory.

Mood Regulation: Tryptophan obtained from oats, bananas, milk, and tuna fishy supports serotonin production, which influences mood in a positive direction.

Social Health:

Cultural Identity: Traditional foods reflect cultural identity, fostering a sense of belonging and community.

Socialization: Shared meals promote communication, strengthening social bonds with family and friends.

Importance of Food

Nutrition and Energy: Food provides the essential nutrients our bodies need for growth, maintenance, and proper functioning. Nutrients such as carbohydrates, proteins, fats, vitamins, and minerals are vital for energy production, tissue repair, immune system support, and various biochemical processes.

Health and Wellness: A balanced and nutritious diet is fundamental for maintaining good health. Adequate intake of nutrients helps prevent nutritional deficiencies, chronic diseases (like heart disease, diabetes), and promotes optimal physical and mental well-being.

Growth and Development: Proper nutrition is particularly important during periods of rapid growth, such as childhood, adolescence, and pregnancy. Nutrients like proteins, calcium, and vitamins are essential for building and maintaining strong bones, muscles, and organs.

Cognitive Function: The brain requires a steady supply of nutrients to function effectively. Nutrients like omega-3 fatty acids, vitamins, and antioxidants support cognitive functions such as memory, concentration, and overall brain health.

Immune System Support: Many nutrients play a crucial role in strengthening the immune system. Adequate intake of vitamins (such as vitamin C and vitamin D) and minerals (such as zinc) helps the body fight off infections and diseases.

Importance of Nutrients

Energy Production: Nutrients such as carbohydrates and fats are primary sources of energy for the body. They provide the fuel needed for daily activities, physical movement, and the functioning of vital organs.

Cell Growth and Repair: Proteins are essential for the growth, repair, and maintenance of cells and tissues. They are crucial for building and repairing muscles, organs, and enzymes.

Vitamin Functions: Vitamins are micronutrients that support a range of biochemical reactions in the body. They are important for processes such as immune system function (vitamin C and D), blood clotting (vitamin K), and antioxidant protection (vitamin E and A).

Mineral Roles: Minerals like calcium, iron, and magnesium are necessary for various bodily functions. Calcium is vital for strong bones and teeth, iron is necessary for oxygen transport in the blood, and magnesium supports muscle and nerve function.

Brain and Nervous System: Nutrients such as omega-3 fatty acids, B vitamins, and certain minerals are essential for proper brain and nervous system function. Omega-3s support cognitive health, B vitamins aid in nerve signalling, and minerals like magnesium help regulate nerve impulses.

Immune System Support: Several nutrients, including vitamins A, C, and D, as well as zinc and selenium, play crucial roles in supporting the immune system. They help the body fight off infections, illnesses, and diseases.

What is Diet

The amount of food that a person consumes in a day is called the diet of the person for the day

BALANCED DIET

"Balanced diet is that food in which all the nutritious elements are present in proper quantity and proportion according to the physical demand, age, weight, activity, etc."

A balanced diet should contain a appropriate amount of Calories, fat, protein, mineral salts, vitamins, and water.

Both excess and deficiency of nutrients are harmful to health.

CHARACTERISTICS OF BALANCED DIET

1. The diet should be sufficient to provide energy

2. Balanced diet should contain all the nutrients in proper proportion.

3. The number of nutrients stored in the body should be high in the diet.

4. A balanced diet should include foods from all the food groups.

5. Food should be attractive, organic/natural, and tasty.

6. A balanced diet typically contains 50-60% carbohydrates, 12-20% protein, and 30% fat

FACTORS AFFECTING BALANCED DIET CHART

The nutritional requirements of a balanced diet vary from person to person. The dietary requirements of individuals of different ages, genders, and body sizes are different. Generally, a balanced diet is affected by the following factors:

1. **Gender: Depending upon the gender the amount of the food would be different and diet plan would be different.**

2. Age : Diet for child, adult and old man would be not be same. Age is an important factors for planning a diet

3. Physical condition: Depending on the physical condition like height, weight, etc. balanced diet can be varied.

4. Amount of Work: Depending on the amount of working **one usually do on daily basis, diet would be different.**

(i) Heavy worker: More amount of food is require

(ii) Moderate worker : Relatively lower energy required, so less food in the diet

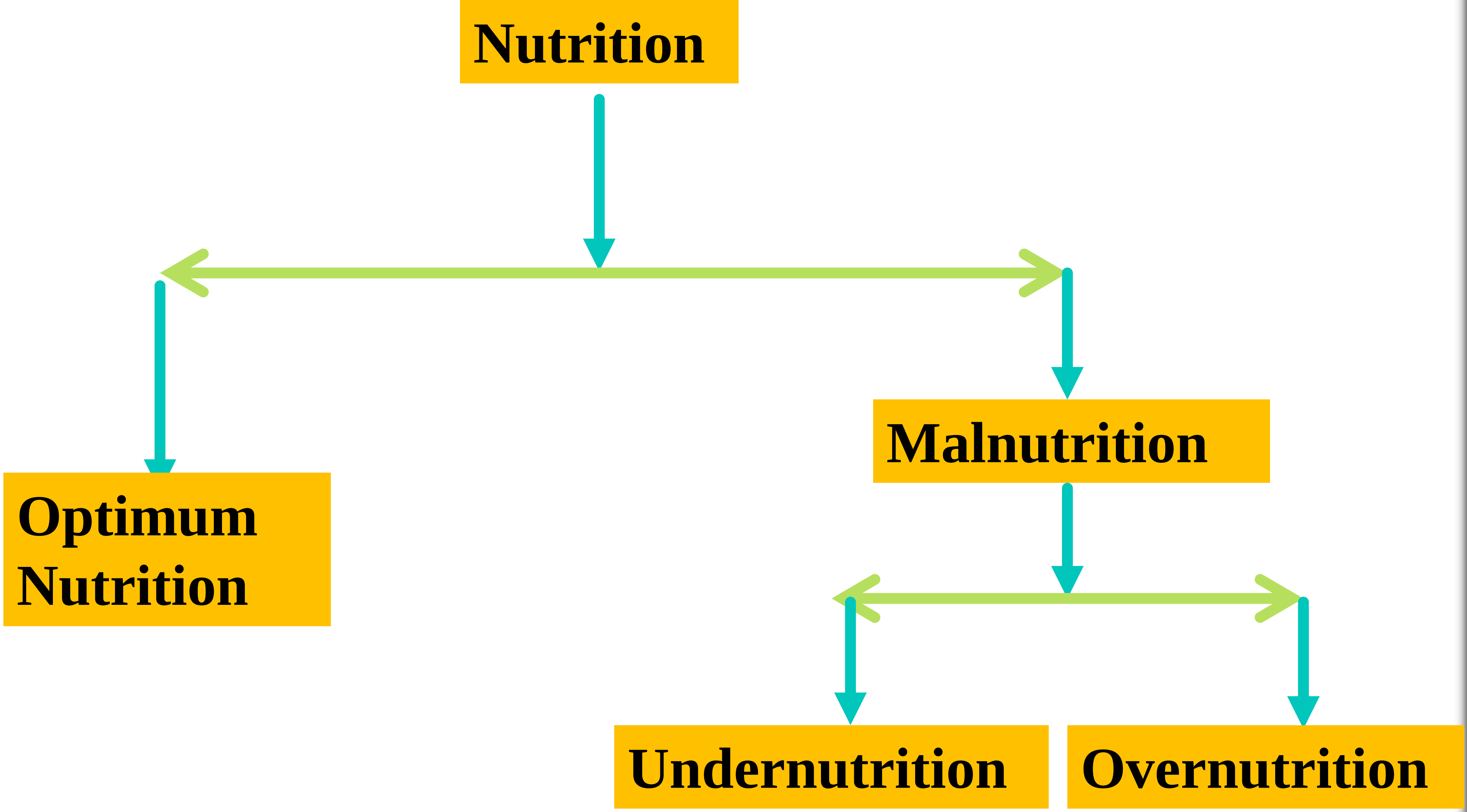
(iii) Sedentary Worker: Table worker require less amount of food

5. Health: Depending on the health condition diet would be different.

6. Climate and Season: Depends on climate, region, geographical origin diet would be different

7. Pregnancy Period: It is the special stage of life and health, needs special care and special diet.

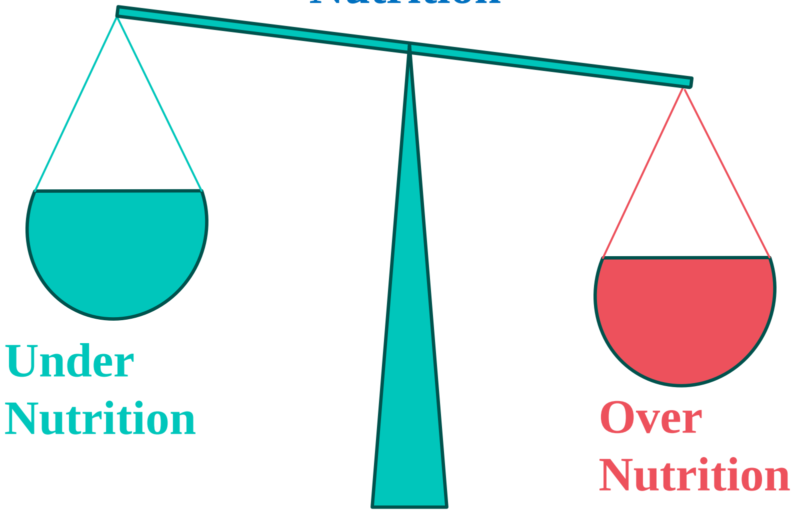
8. Lactation Period: In this period mother mammal produce milk and feed to the baby mammals. At this time diet should be different from the average time


Nutrition
Optimum Nutrition
Malnutrition
Undernutrition
Overnutrition

OPTIMUM NUTRITION

Optimum nutrition is also called best nutrition. It can be defined as taking the right amount of nutrients on proper schedule to achieve the best performance and the longest lifetime in good health.

Undernutrition: Deficiency in nutrients or energy



Overnutrition: Excess in energy and nutrients
It may causes (diabatic, high pressure, stroke)

MALNUTRITION

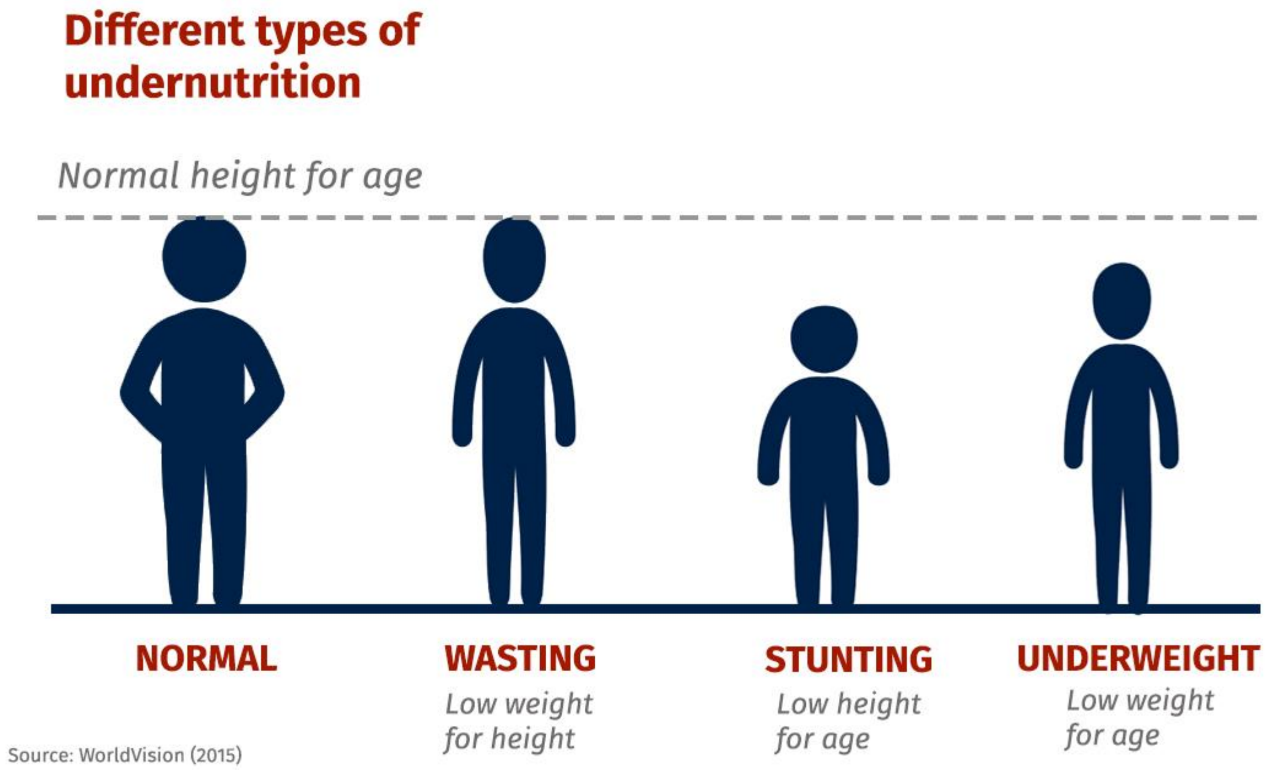
Malnutrition is a condition that results from eating diets for long time in which one or more nutrients are either not enough or present in excess and this diet causes health problems. This health condition called malnutrition.

Impact of Under Nutrition

Wasting: Low weight with respect to height.

Stunting: Shorter height with respect to age

Underweight: Low weight with respect to age



Impact of Over Nutrition

- **Obesity**: A disorder which occurs by eating more calories than one burns
- **Oversupplying a specific nutrient**, such as dietary, minerals, or vitamin poisoning.

For mineral excess: Iron poisoning (vomiting, Liver damage)

Diabetes Hypertension, Cardiovascular disease, and stroke

Characteristics of the optimum nutrition-The following are the characteristics of optimum nutrition:

1. The body of a person receiving optimum nutrition remains strong and there is a balance in his body length proportion.
2. The body’s digestive organs, blood circulation, and respiratory system function well. The muscles of a well-nourished person are healthy and strong.
3. There is a normal layer of fat in the body which makes the skin look smooth and healthy.
4. Teeth and gums are healthy. Hair is long, shiny, and thick. Eyesight is correct and their vision is perfect.
5. He always feels fresh, he feels normal hunger, and he is less sick. He can work hard and does not feel tired soon because of his healthy stamina..

The Importance of optimum Nutrition-Most people know that optimum nutrition and physical activity help maintain a healthy weight, but the benefits of optimum are more than that. The importance of optimum nutrition can be explained under the following points:

1. **Reduces the risk of certain diseases including heart disease, diabetes, stroke etc.**
2. **Helps in reducing high blood pressure.**
3. **Helpful in reducing cholesterol.**
4. **Helps in staying healthy.**
5. **Enhances the ability to fight disease and recover from injury.**
6. **Helpful in increasing energy levels.**
7. **Helps for proper growth and development**

Symptoms of Inadequate Nutrition—The main symptoms of insufficient nutrition are:

- The skin becomes wrinkled, dry, and lifeless.
- The body weight gradually decreases.
- Hair loses its shiny property and looks dry.
- The body appears yellow due to a lack of blood.
- The eyesight decreases, and night blindness and other eye diseases occur.
- The bones of children get deformed and their growth slows down.
- There is no interest in doing any work and the ability to work decreases, fatigue is felt quickly. There is a lack of physical and mental strength.
- Due to the low immunity of the body, a person becomes easily affected by infectious diseases.
- Imperfectly developed teeth and erratic eruption of teeth in the jaws are symptoms of inadequate nutrition.

Malnutrition

- Affects 1 in 3 people all over the world
- Comes in different shapes and sizes

Stunting	Wasting	Obesity
A person too short for his age	A person too thin for his height	A person who's overweight

Image source : Rappler.com

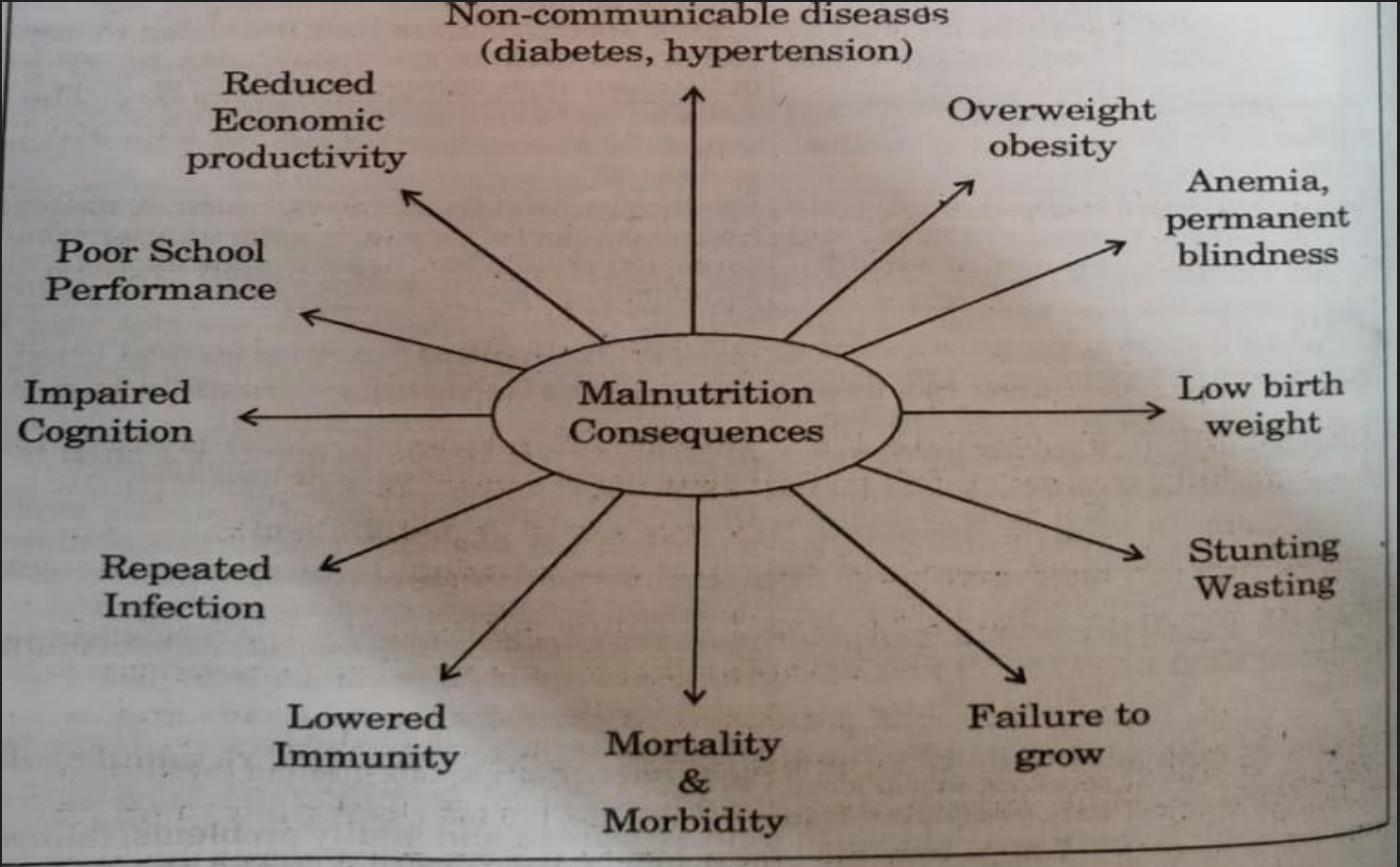


Fig. 2 : Consequences of malnutrition

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Causes of Malnutrition

Poverty-Poverty is plentiful in those countries where this problem is there. Economically week family unable to arrange proper food. This is one of the reason behind malnutrition.

Illiteracy/Unawareness -The unawareness of proper nutrition leads most of the people to the stage of malnutrition.

Shortage of food items- Different types of food are not available in every parts of country. Less supply of food in different part of country may also causes for malnutrition.

Adulteration in food items-Today a corrupt way of earning money is by adulteration of food items like adding stones in papaya seeds and in black pepper, mixing of flour in turmeric, mixing of urea in milk etc.

Wrong food habits-Sometimes we continue with our wrong food habits despite our education and knowledge. Even after knowing that too much chili, spices, and fried greasy food are harmful to us, due to the good taste, we are not able to leave them. The wrong time of eating and the increased gap between two meals is also included in our wrong habits.

Unhealthy environment–Lack of clean air, lack of sunlight and lack of cleanliness can also be the reason for malnutrition.

Plan to reduce malnutrition is express as follows (any three points)

Awareness camp: We can arrange an awareness camp to enhance awareness of nutrition and especially malnutrition because the most important reason behind malnutrition is unawareness of proper food, balanced diet and nutrition.

Make available different schemes: There are different government schemes available that provide food/grain free of cost to the general people. We can make sure that everyone gets benefits from these schemes, which will help to distribute food to the needy and reduce malnutrition.

New farming Techniques: There are several techniques (high-quality seed, management of fertilizer, soil testing and farm suitable crops) that increase food and crop production. By learning about the techniques crop and food production can be increased which helps to reduce malnutrition.

Improving Sanitation and Clean Water Access: We can improve sanitation and cleanliness in our families and surroundings. Cleanliness and sanitation play a very crucial role in reducing malnutrition.
Fundraising:

Fundraising: We can make a food bank in our locality where leftover food from every house will be submitted and from where the needy can take it as per their requirement. That also helps to reduce malnutrition at the ground level.

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Prevention/treatment of Malnutrition

Proper awareness,

Meal/Diet with sufficient nutrients,

Appropriate schedule of food intake

If necessary, can take supplementary nutrient after consulting with doctor:

Proper exercise :

Improving Access to Nutritious Food:

Promoting Dietary Diversity:.

Enhancing Maternal and Child Nutrition:

Improving Sanitation and Clean Water Access:

Strengthening Healthcare Services:

Education and Behaviour Change:

Social Safety Nets.

Agricultural and Food Production Interventions:

Monitoring and Data Collection:

Legal and Policy Framework:

Carbohydrates ⟷ **4 cal/g**

Protein ⟷ **4 cal/g**

Fat ⟷ **9 cal/g**

Vitamins and minerals ⟷ **0 cal/g**

Height in Cm	Weight in KG (Age Range in Years)			
	Age 15-17	18-22	23-27	28-32
152 – 158	46 – 49	47 – 50	50 – 54	54 – 58
159 – 165	50 – 53	51 – 55	55 – 59	59 – 63
166 – 171	54 – 56	56 – 59	60 – 64	63 – 66
172 – 178	57 – 60	59 – 63	64 – 69	67 – 71
179 – 183	61 – 63	64 – 66	69 – 72	72 – 74
184 – 185	64	67 – 68	73 – 74	75

height and weight required for Indian army for Women

Height in Cm	Weight in KG (Age Range in Years)	
	Age 20-25	26-30
148 – 151	43 – 45	46 – 48
152 – 155	46 – 48	49 – 51
156 – 160	49 – 51	52 – 55
161 – 165	52 – 54	55 – 58
166 – 171	55 – 58	59 – 62
172 – 176	59 – 61	63 – 66
177 – 178	62 – 63	

BMI (body mass index): $\frac{Weight\ (kg)}{[Height(m)]^2}$

Men: BMR = 88.362 + (13.397 x weight in kg) + (4.799 x height in cm) – (5.677 x age in years)

- **Women: BMR = 447.593 + (9.247 x weight in kg) + (3.098 x height in cm) – (4.330 x age in years)**

BMR: Basal Metabolic Rate: It is the number of calories your body burns during complete rest

BMR (Mine) =88.362 +(13.397 x 68) + (4.799 x 169) – (5.677 x 31)
=1634 cal/day

BMR
= 1634 Cal/day

Activity Level	Calorie
Sedentary: little or no exercise	1,961 ~300
Exercise 1-3 times/week	2,247 ~600
Exercise 4-5 times/week	2,394 ~750
Daily exercise or intense exercise 3-4 times/week	2,533
Intense exercise 6-7 times/week	2,819
Very intense exercise daily, or physical job	3,105

Men: BMR = 88.362 + (13.397 x weight in kg) + (4.799 x height in cm) – (5.677 x age in years)

= 88.36+(13.39 x 68 kg) +(4.799 x 169 cm) – (5.677 x 32 in year)

- Women: BMR = 447.593 + (9.247 x weight in kg) + (3.098 x height in cm) – (4.330 x age in years)

Principle of Meal or diet planning/ point has to be in mind before meal planning

1. **Family-friendly food**
2. **Psychosis and Nutrition-**
3. **Nutritional Requirements-**
4. **Nutrition and mental strength-**
5. **Food served in an attractive manner-**
6. **Innovative diet-**

(i) Novelty in colour scheme-

(ii) Use of Cooking Methods–

(iii) Selection of food items from different food groups-

(iv) Novelty in texture—

(v) Novelty in taste and aroma-

7. **According to the budget of the family**
8. **Saving time, power, and money-**
9. **Use of leftover food material-**
10. **Satiety**.

	Low Labor	Moderate Labor	High Labor
Total Energy Calories	2425	2875	3800
Protein (in grams)	60	60	60
Fat (in grams)	20	20	20
Calcium (in mg)	400	400	400
Iron (in mg)	28	28	28
Retinol (μg)	600	600	600
Carotene (μg)	2400	2400	2400
Thiamine (mg)	1.2	1.4	1.6
Riboflabin (mg)	1.4	1.6	1.9
Niacin (mg)	16	18	21
Ascorbic Acid (mg)	40	40	40
Folic Acid (μg)	100	100	100
Vitamin B_{12}, (μ) min. Gram	1	1	1

Proposed daily intake of nutrients for women

	Low labor	*Moderate labor*	*High labor*	*Pregnant women*	*Lactation*	*0-6 months*
Total energy (kg) calories	1875	2225	2925	+300	+550	
Protein (grams)	50	50	50	+15	+25	
Fat (grams)	20	20	20	30	45	
Calcium (mg)	400	400	400	1000	1000	
Iron Elements (mg)	30	30	30	38	30	
Retinol (μg)	600	600	600	600	250	
Carotene (μg)	2400	2400	2400	2400	3800	
Thiamine (mg)	0.9	1.1	1.2	0.2	0.3	
Riboflavin (mg)	1.1	1.8	1.5	0.2	0.5	
Niacin (mg)	12	14	16	2	4	
Ascorbic Acid (mg)	40	40	40	40	80	
Folic Acid (μg)	100	100	100	400	450	
Vitamin B_{12} (μ) min.gram	1	1	1	1	1	

Recommended daily intake of nutrients for infant and child

Nutrients	*Infant*				
	0-6 months	*6-12 months*	*1-3 years*	*4-6 years*	*7-9 Years*
Total Energy Calories	108/kg	98/k	1240	1690	1950
Protein (in grams)	2.05/kg	1.65/kg	22	30	41
Fat (in grams)	—	—	25	25	25
Calcium (mg)	500	500	4010	400	400
Iron content (mg)	—	—	12	18	16
Retinol (μg)	350	350	400	400	600
Carotene (μg)	1200	1200	1600	1600	2400
Thiamine (mg)	155μgm/1kg	55μg/min.	0.6	0.9	1.0
Riboflavin (mg)	65ug/1 kg	60mg	0.7	1.0	1.2
Niacin (mg)	1710μgm/kg	650μg/1(kg)	8	11	12
Ascorbic Acid (mg)	25	25	40	40	60
Folic Acid (μg)	25	25	30	40	60
Vitamins (μ),min.gram	0.2	0.2	0.2-1.0	0.2-1.0	0.2-1.0

Nutrients Available in Foods Nutrients in Food Classes

Main Nutrients Proposed by ICMR

Food Group	*Main Nutrients*
I. **Cereals, Grains and Products :** Rice, Wheat, Ragi, Bajra, Maize, Jowar, Barley, Rice flakes, Wheat Flour.	Energy, protein, Invisible fat Vitamin B1, Vitamin–B2, Folic Acid, Iron, Fibre.
II. **Pulses and Legumes :** Bengal gram, Black gram, Green gram, Red gram, Lentil (whole as well as dhals) Cowpea, Peas, Rajmah, Soyabeans, Beans	Energy, Protein, Invisible fat, Vitamin – B1, Vitamin – B2, Folic Acid, Calcium, Iron, Fibre.
III. **Milk and Meat Products :** Milk: Milk, Curd, Skimmed milk,Cheese Meat: Chicken, Liver, Fish, Egg, Meat.	Protein, Fat, Vitamin –B12, Calcium. Protein, Fat, Vitamin –B2
IV. **Fruits and Vegetables :** **Fruits**: Mango, Guava, Tomato Ripe, Papaya, Orange. Sweet Lime, Watermelon.	Carotenoids, Vitamin –C, Fibre.
Vegetables (Green Leafy) : Amaranth, Spinach, Drumstick leaves, Coriander leaves, Mustard leaves, fenugreek leaves	Invisible Fats, Carotenoids, Vitamin – B2, Folic Acid, Calcium, Iron, Fibre.
Other Vegetables : Carrots, Brinjal, Ladies fingers, Capsicum, Beans, Onion, Drumstick, Cauliflower.	Carotenoids, Folic Acid, Calcium, Fibre

	protein g/100g		carbohydrate g/100g		fat g/100g		thiamine (B1)	
	new	old	new	old	new	old	new	old
bajra	10.96±0.26	11.6	61.78±0.85	67.5	5.43±0.64	5	0.25±0.044	0.33
rice (raw milled)	7.94±0.58	6.8	78.24±1.07	78.2	0.52±0.05	0.5	0.05±0.019	0.06
wheat (whole)	10.59±0.60	12.8	64.72±1.74	71.2	1.47±0.05	1.5	0.46±0.067	0.45
red gram dal	21.7±0.5	22.3	55.23±0.83	57.6	1.56±0.03	1.7	0.45±0.046	0.45
green gram whole	22.53±0.43	24	46.13±0.64	56.7	1.14±0.17	1.3	0.45±0.027	0.47
lentil brown whole	22.49±0.58	25.1	48.47±1.12	59	0.64±0.02	0.7	0.40±0.073	0.45
cabbage green	1.36±0.07	1.8	3.25±0.91	4.6	0.12±0.01	0.1	0.03±0.005	0.06
brinjal (all varitie	1.48±0.22	1.4	3.52±0.80	4	0.32±0.04	0.3	0.06±0.016	0.04
tomato green	1.12±0.08	1.9	3.18±0.4	3.6	0.27±0.07	0.2	0.08±0.043	0.07
tomato (ripe, hybrid)	0.76±0.03	0.9	3.20±0.30	3.6	0.25±0.02	0.2	0.04±0.004	0.12
potato (brown skin, big)	1.54±0.17	1.6	14.89±0.40	22.6	0.23±0.02	0.1	0.06±0.004	0.1
apple	0.29±0.08	0.2	13.11±0.76	13.4	0.64±0.04	0.5	0.03±0.017	n/a
banana (ripe, robusta)	1.23±0.08	1.2	23.63±0.74	27.2	0.33±0.01	0.3	0.01±0.0	0.05
papaya	0.42±0.05	0.6	4.61±0.48	7.2	0.16±0.01	0.1	0.03±0.009	0.04
mustard seeds	19.51±0.23	20	16.80±0.71	23.8	40.19±0.21	39.7	0.55±0.065	0.65
groundnut	23.65±0.85	25.3	17.27±0.33	26.1	39.63±0.29	40.1	0.57±0.052	0.9
milk buffalo	3.68±0.13	4.3	8.39±0.71	5	6.58±0.20	6.5	0.05±0.006	0.04
milk cow	3.26±0.06	3.2	4.94±1.02	4.4	4.48±0.29	4.1	0.03±0.006	0.05
eggs (poultry, whole)	13.28±0.29	13.3	n/a	n/a	9.15±0.04	13.3	0.06±0.01	0.1
Rohu	19.71±0.57	16.6	n/a	4.4	2.39±0.64	0.9	n/a	0.05

Table : Balanced diet food items for an adult male

Food (in grams)	*Sedentary Worker*		*Moderate Worker*		*Heavy Worker*	
	Veget-arian	Non veget-arian	Veget arian	Non veget arian	Veget-arian	Non veget arian
Grain	350	350	425	425	600	600
Cereals	70	55	80	65	80	65
Green Leafy Vegetables	100	100	125	125	125	125
Other Vegetables	75	75	100	100	100	100
Roots and Tubers Vegetables	75	75	100	100	100	100
Fruits	60	60	60	60	60	60
Milk	600	400	600	400	600	400
Moss, Fish	—	60	—	60	—	60
Egg	—	30	—	30	—	30
Jaggery and Sugar	30	30	40	40	55	55
Groundnut	50	50	50	50	50	50
Ghee and Oil	35	40	40	40	50	50

Table : Balanced Diet for Normal Adult Female, Pregnant Mothe and Nursing Mother

Food Items (in grams)	*Sedentary Worker*		*Moderate Worker*		*Heavy Worker*	
	V	NV	V	NV	V	NV
Cereal	260	250	310	300	440	425
Lentils	60	50	60	50	60	50
Green Leafy Vegetables	100	100	100	100	100	100
Other vegetables	75	75	75	75	100	100
Root and tuber vegetables	50	50	75	75	100	100
Fruits	60	60	60	60	60	60
Milk	400	250	250	400	400	250
Eggs	—	30	30	—	—	30
Meat, fish	—	60	60	—	—	60
Ghee and Oil	30	35	40	40	40	45
Jaggery	30	30	30	40	40	40
Groundnut	50	30	30	50	50	30

Sugars:
Sugar, Jaggery

SAMPLE BALANCED MEAL PLAN

Meal Timing	*Food Items*	*Household Measures*
Breakfast	Vegetable poha	1 bowl
	Milk	1 glass
	Paneer/egg/sprouts	1 serving/1 tbsp
Mid Morning	Fruit	1
Lunch	Salad	1 Plate
	Chapati or Rice	3 ladle
	Dal/Sambhar	1 bowl
	Cauliflower potato	1 bowl
	Curd	1 bowl
	Mint Coriander Chutney	1 tbsp
Evening	Tea/Milk	1 cup
	Sprouts/Upma/roasted gram	1 bowl/25 g
Dinner	Tomato Soup/Rasam	1 cup
	Chapati of rice flour	3 ladle
	Masoor dal	1 bowl
	Green vegetables	1 bowl
	Sweet dish	1 bowl

CLASSIFICATION OF FOOD (FOOD GROUPS)

On the basis of origin

Plant origin: Vegetables, fruits and grain

Animal Origin: Milk, Meet, Eggs

Mineral: Mineral salt, Rock Salt

On the basis of Chemical Composition/Nutrients

Carbohydrates,

Protein, Fats

Vitamin, mineral, Fibre, Water

On the basis of Function

Energy source Food: Carbohydrates, protein, fats

Body building Food: fats and proteins, minerals

Protective Food: High vitamin, mineral

CLASSIFICATION OF FOOD ON THE BASIS OF NUTRIENTS or CHEMICAL COMPOSITION

On the basis of the above nutrients, the food items are divided into the following categories-

Foods containing carbs-There are two nutrients under carbs (i) sugar, and (ii) starch. Sugary food items include sugar, jaggery, honey, sweets, marmalade, and jam-jelly, etc, and starchy foods include rice, sago, wheat, maize, jowar, millet, sweet potato, etc.

Fat-rich food items-Ghee, fat, butter, cream, vegetable oil, groundnut, oilseeds, dry fruits, etc. come under the food items containing fat.

Protein-rich foods-We gets proteins from animals and plants. A protein-rich food is found in all types of meat, fish, liver, eggs, milk, milk products, pulses, soybeans, nuts, groundnuts, etc.

Foods containing mineral salts-Meat, fish, eggs, liver, and milk are found in all cereals, pulses, green leafy vegetables, etc., under the category of food items containing mineral salts.

Vitamin-rich foods-Almost all green vegetables and other foods have some or the other vitamin. But milk and its products, meat, fish, liver, egg, green vegetables, etc., contain vitamins that are found in a very large quantity.

CLASSIFICATION OF FOOD ACCORDING TO FUNCTIONS

The types of food have been classified as follows:

(A) Body-building food elements- The food elements that work to build the body are called the body-building elements. Protein is the main component of body-building food. Protein helps us to grow and repair our wear and tear. Every organ such as bones, muscles, teeth, skin, hair, and blood is made up of proteins. Fats also help to build body organ

(B) Energy sources or energy-providing substances-Energy is essential for the body. Different bodily functions require different amounts of energy.

Carbohydrate is the main food providing energy in the body. Sugar and starch are the two forms of carbohydrates available in the body.

(1) Sugars—Sugar, Honey, Jaggery, etc.

(ii) Starch-Various grains such as wheat flour, semolina, maida (flour), and maize. barley, millet, etc. Apart from this, starch content is found in sweet potato, potato, raw fruit, and banana.

Fat- 1 gram of fat provides 9 calories of energy. We can get fat from other food items like oil, ghee, butter, nuts, cream, etc. Hardworking people should take more amount of fat in their diet.

Protein—Most of the energy is supplied from carbs as it is found in more food items and is digestible. Protein also serves as a source of energy. 4 calories of energy are obtained in one gram of protein. The main function of protein is the repair and regulation of body's tissues. Carbohydrates and fats also help in protein utilization and metabolism. This allows the protein to perform its main function properly.

We get protein from milk, curd, cheese, meat, fish, eggs, pulses, soybeans, dry fruits, peanuts, etc.

(C) Regulatory or protective food elements—Along with energy-giving elements and building elements, our body also needs protective elements. The biological activities of our body are controlled by protective elements so that the organs function properly. Protective elements are necessary to increase the immunity power of the body.

Vitamins, minerals, water, and proteins are the main protective elements. Vitamins regulate various body functions and keep organs healthy. Vitamin A especially keeps the skin and eyes healthy.

CLASSIFICATION OF FOOD GROUPS ON THE BASES OF SOURCES

Plant Origin

Cereals

Cereals and millets constitute by far the most important group of foodstuffs as they form the staple food of a large majority of the population throughout the world. They form about 70 to 80 percent of the diet of the low-income groups in India and other developing countries. They contain about 6-12 percent proteins and are good sources of some B vitamins e.g. thiamine, niacin, and vitamin B, and minerals e.g. phosphorus and iron.

Hence, they provide about 70 to 80 percent of the calories, proteins, and other nutrients mentioned above in the diets of the low-income groups. All cereals except ragi are poor to moderate sources of calcium. Ragi is one of the richest sources of calcium, containing about 0.4 percent calcium. Cereals are deficient in vitamins A, D, B_{12}, and C. Yellow maize, however, contains fair amounts of vitamin A. Puffed cereals are consumed widely as a snack by low-income groups in India.

Pulses

Dried pulses are rich in proteins containing about 19-24 percent. They are good sources of many B vitamins and minerals but are **deficient in vitamins A, D, B_{12}, and C**. They effectively supplement cereals. Puffed pulses, e.g. puffed Bengal gram and tapioca are the main contributory cause for the wide prevalence of protein-calorie malnutrition in some countries in Africa, Asia, and Latin America, especially among children.

Other vegetables-This group includes a large number of vegetables. Some of them are good sources of vitamin C. Yellow pumpkin is a fair source of carotene.

Fruits

Fruits, in general, are good sources of **vitamin C**. Some of them e.g. mango and papaya, are fair sources of carotene, and Indian gooseberry (**Amla**) and guava are very rich sources of vitamin C. They are also the cheapest fruits. Other fruits which are good sources of vitamin C are tomatoes, citrus fruits, papaya, cashew fruit, and pineapple. Apple, bananas, are poor sources of vitamin C.

Animal Sources

Milk and milk products

Milk has always been used as the staple food for infants and as a supplement to the diets of children and adults. Milk is almost a complete food except for **deficiencies of iron and vitamins C and D.** Milk proteins are of high biological value. One litre of cow's milk provides about 35 g protein, 35 g fat, 1 g calcium, 1.5 mg riboflavin, 1500 I.U. of vitamin A, and substantial amounts of other **B vitamins and minerals**. Buffalo milk is used extensively in India. Its fat content is about twice that of cow's milk.

Full-fat milk powder: Full-fat milk powder is about 8 times as rich as cow's milk containing about 26 percent proteins and 26 percent fat. It can be reconstituted to 7 times its weight with warm water and used in place of fresh milk.

Skimmed milk powder: Skim milk powder is prepared from fat-free milk. It is completely devoid of fat and vitamin A. It is about 10 times as rich as fresh skim milk and contains about 35 percent proteins. It can be used as a supplement to the diets of children. **It is not suitable for feeding infants.**

Eggs

Hen's egg contains about **13 percent proteins** of very high biological value and **13 percent fat.** It is a rich source of vitamin A and some B vitamins. It is a fair source **of vitamin D** but does **not contain vitamin C**. Egg white contains about 12 percent proteins and some B vitamins but is devoid of fat and vitamin A. **Egg yolk contains about 15 percent proteins and 27 percent fat**. It is a rich source of vitamin A and a fair source of iron, B vitamins, and vitamin D, and therefore it is used as a supplement for infants.

Meat, fish, and other animal foods

Meat: Meat is rich in proteins (18-22 percent) of high biological value. It is a fair source of B vitamins. It does not contain any vitamins A, C, or D.

Fish: Fish is rich in proteins (18-22 percent) of high biological value. It is a fair source of B vitamins. Fatty fish contain some vitamins A and D. Large fish are rich in phosphorus but are deficient in calcium. Small fish eaten with bones are good sources of calcium.

Liver: Liver is rich in proteins (18-20 percent) and vitamin A and B-complex. It is the richest natural source of vitamin B_{12}.

Fats and Oils

Fats and oils serve mainly as a source of energy and provide essential fatty acids. Butter and ghee and Vanaspathi are good sources of vitamin A (about 2500 I.U. per 100 g). The common vegetable oils and fats do not contain carotene or vitamin A. Many of them are good sources of vitamin E.

Sugar and other carbohydrate foods

The carbohydrate foods commonly used are cane sugar, jaggery, glucose, honey, syrup, custard powder, arrowroot flour, and sago. They serve mainly as a source of energy. Honey and jaggery contain small quantities of minerals and vitamins.

Condiments and spices

Condiments and spices are not important sources of nutrients in average diets but are used mainly for enhancing the palatability of the diet. The essential oils present in them help to improve the flavour and acceptability of food preparations. Some time its improve immunity of the body

Concept of Food and Nutrition

(a) Definition of Food, Nutrients, Nutrition, Health, balanced Diet

(b) Types of Nutrition- Optimum Nutrition, under Nutrition, Over Nutrition

(c) Meal planning- Concept and factors affecting Meal Planning

(d) Food groups and functions of food

Nutrients: Macro and Micro

RDA, Sources, Functions, Deficiency and excess of

(a) Carbohydrate, (b) Fats, (c) Protein, (d) Minerals

Major: Calcium, Phosphorus, Sodium, Potassium

Trace: Iron, Iodine, Fluorine, Zinc

(e) Vitamins: Water soluble vitamins: Vitamin B, C

Fat soluble vitamins: Vitamin A, D, E, K

(f) Water, (g) Dietary Fiber

RDA- Recommended Dietary Allowances

SUMMARY OF RDA FOR INDIANS – 2020

Age Group	Category of work	Body Wt (kg)	Protein (g/d)	CHO (g/d)	Calcium (mg/d)	Magnesium (mg/d)	Iron (mg/d)	Zinc (mg/d)	Iodine (µg/day)	Thiamine (mg/d)	Riboflavin (mg/d)	Niacin (mg/d)	Vit B6 (mg/d)	Folate (µg/d)	Vit B12 (µg/d)	Vit C (mg/d)	Vit A (µg/d)	Vit D (IU/d)
Men	Sedentary									1.4	2.0	14	1.9					
	Moderate	65	54.0	130	1000	385	19	17	150	1.8	2.5	18	2.4	300	2.5	80	1000	600
	Heavy									2.3	3.2	23	3.1					
Women	Sedentary									1.4	1.9	11	1.9					
	Moderate	55	45.7	130	1000	325	29	13.2	150	1.7	2.4	14	1.9	220	2.5	65	840	600
	Heavy									2.2	3.1	18	2.4					
	Pregnant woman	55 + 10	+9.5 (2nd trimester) +22.0 (3rd trimester)	175	1000	385	40	14.5	250	2.0	2.7	+2.5	2.3	570	+0.25	+15	900	600
	Lactation 0-6m		+16.9	200	1200	325	23	14	280	2.1	3.0	+5	+0.26	330	+1.0	+50	950	600
	7-12m		+13.2	200						2.1	2.9	+5	+0.17	330				
Infants	0-6 m*	5.8	8.1	55	300	30	-	-	100	0.2	0.4	2	0.1	25	1.2	20	350	400
	6-12m	8.5	10.5	95	300	75	3	2.5	130	0.4	0.6	5	0.6	85	1.2	27	350	400
Children	1-3y	11.7	11.3	130	500	135	8	3.0	90	0.7	0.9	7	0.9	110	1.2	27	390	
	4-6y	18.3	15.9	130	550	155	11	4.5	120	0.9	1.3	9	1.2	135	1.2	32	510	600
	7-9 y	25.3	23.3	130	650	215	15	5.9	120	1.1	1.6	11	1.5	170	2.5	43	630	
Boys	10-12y	34.9	31.8	130	850	270	16	8.5	150	1.5	2.1	15	2.0	220	2.5	54	770	600
Girls	10-12y	36.4	32.8	130	850	255	28	8.5	150	1.4	1.9	14	1.9	225	2.5	52	790	600
Boys	13-15y	50.5	44.9	130	1000	355	22	14.3	150	1.9	2.7	19	2.6	285	2.5	72	930	600
Girls	13-15y	49.6	43.2	130	1000	325	30	12.8	150	1.6	2.2	16	2.2	245	2.5	66	890	600
Boys	16-18y	64.4	55.4	130	1050	405	26	17.6	150	2.2	3.1	22	3.0	340	2.5	82	1000	600
Girls	16-18y	55.7	46.2	130	1050	335	32	14.2	150	1.7	2.3	17	2.3	270	2.5	68	860	600

Unit-II

Nutrient

Nutrients that are required in vast amounts are called macronutrients. There are three types of macronutrients:

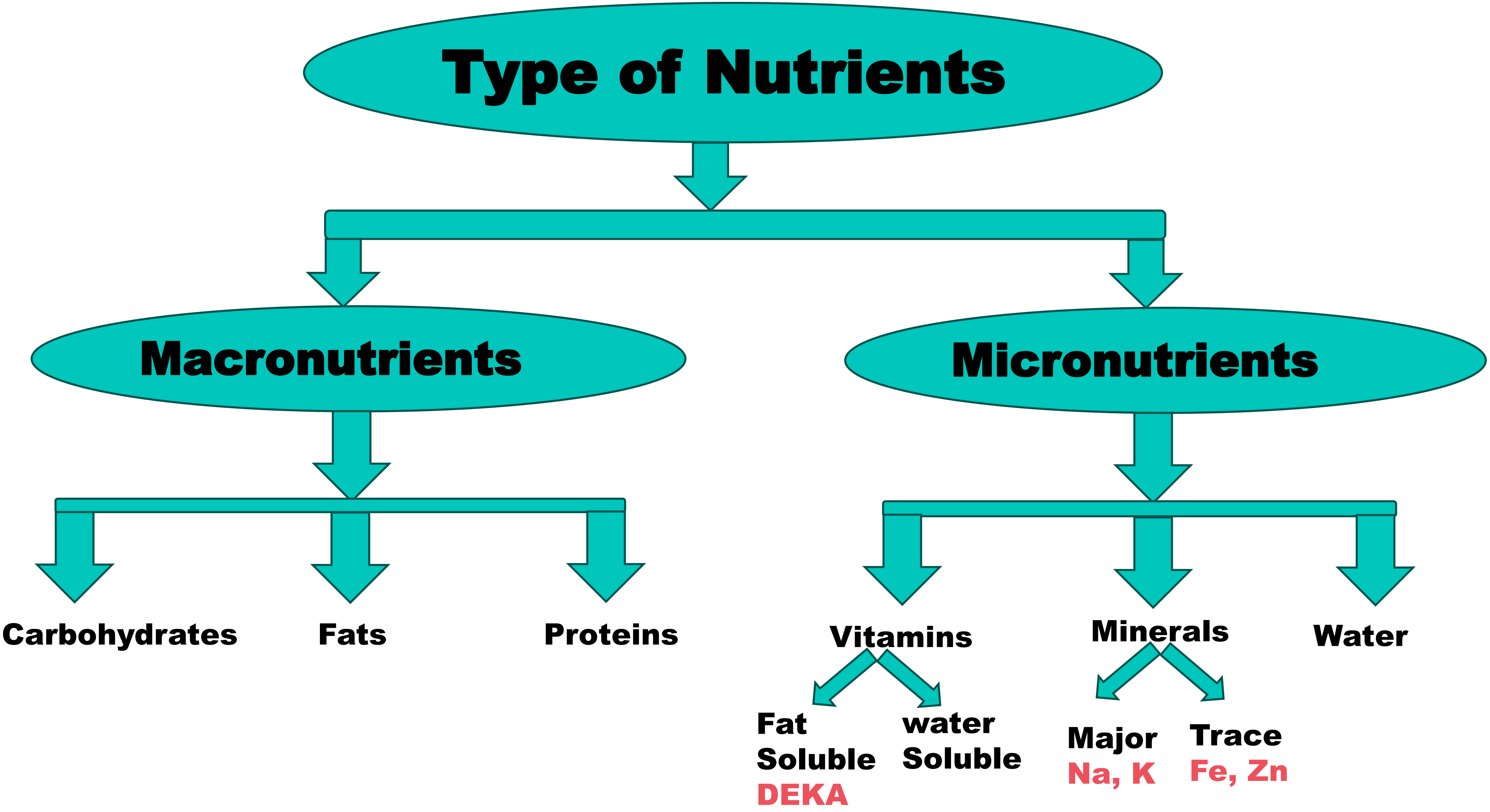
Carbohydrate, protein, fat

Macronutrients are mostly carbon-based compounds that can be metabolically processed into cellular energy through changes in their chemical bonds. The chemical energy is transformed into cellular energy known as ATP, which is used by the body to carry out basic functions.

The quantity of energy a person consumes on an everyday basis originates basically from the 3 macronutrients carbohydrates, protein, fat. Food energy is measured in calories or kilocalories.

Nutrients that are required in small amount are called micronutrients.

Vitamin, mineral, water


Type of Nutrients
Macronutrients
Micronutrients
Carbohydrates
Fats
Proteins
Vitamins
Minerals
Water
Fat Soluble
DEKA
water Soluble
Major
Na, K
Trace
Fe, Zn

Protein

Proteins were first described by the Dutch chemist Gerardus Johannes Mulder and named by the Swedish chemist ***Jöns Jacob Berzelius*** in 1838.

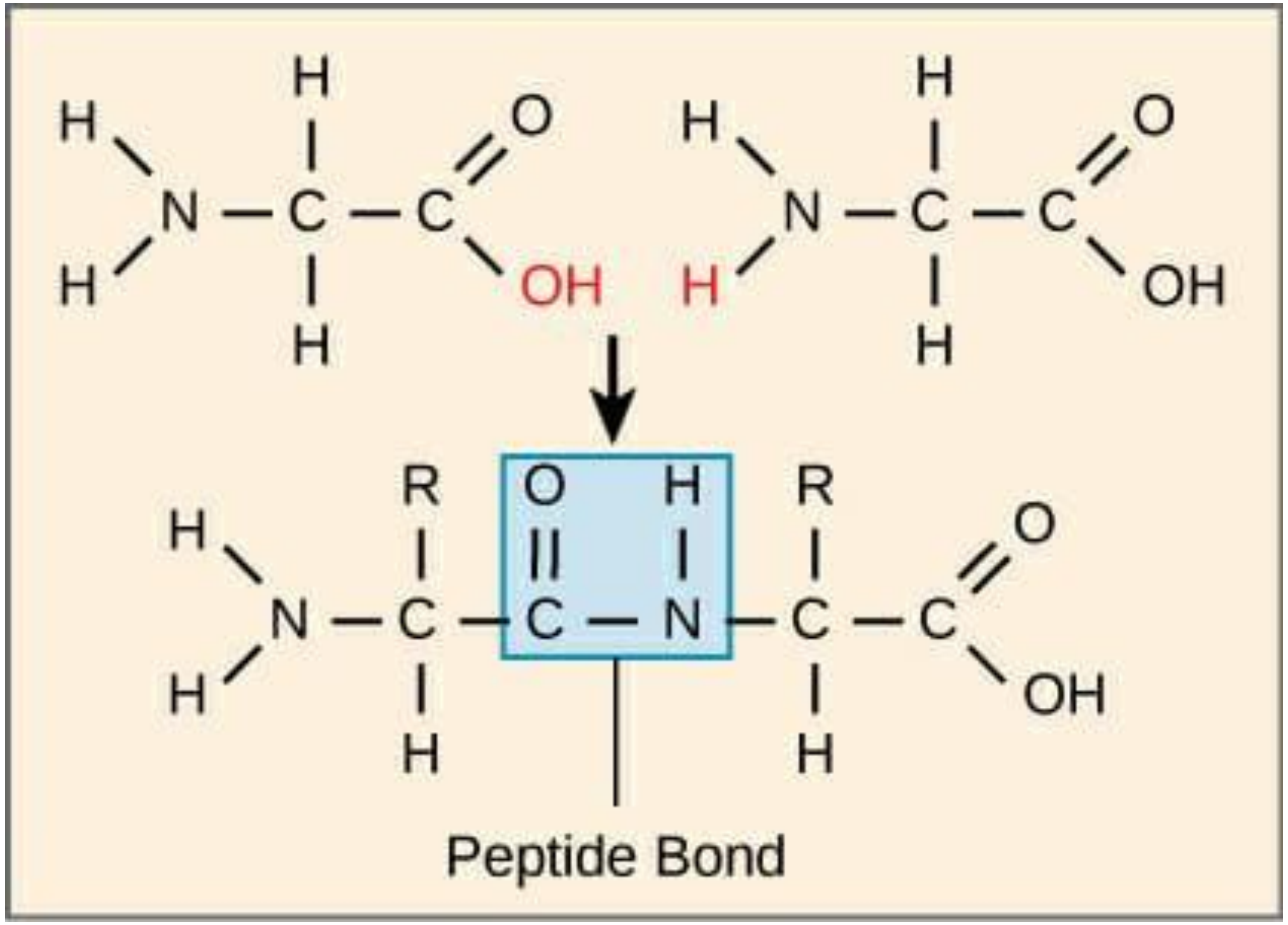
Chemical Organization- Protein is an organic compound. It contains carbon, hydrogen, oxygen, and nitrogen. Apart from these, sulphur and phosphorus are also present in some protein molecules. Iron, copper, cobalt, and other salts are also present along with some protein molecules. Nitrogen is the main component of protein.

Amino Acids—Protein breaks down into its simplest unit amino acids after hydrolysis. Therefore, the amino acid is the smallest unit of protein. Many amino acids combine to form proteins.

Protein molecules are very large. Their molecular weight ranges from 15,000 – 50,000.

Definition-Proteins are the main constituent of all the body cells and thus form the greater part of muscles and other tissues.

A total of 23 amino acids are found in the food substance, and more than 120 amino acids in the protein are joined by a peptide linkage to form a protein molecule.



Classification of proteins

Proteins are large molecules formed by the combination of a number of amino acids that have been found to occur in proteins and are important from the point of view of human nutrition. Amino acids can be classified as follows:

Essential	*Semi-essential*	*Non-essential*
Histidine	Arginine	Glutamic acid
Lysine	Tyrosine	Aspartic acid
Tryptophan	Cystine	Alanine
Phenylalanine	Glycine	Proline
Methionine	Serine	Hydroxyproline
Threonine		Cysteine
Leucine		
Isoleucine		
Valine		

Classification of Proteins (Based on chemical composition)

Simple proteins: It is composed entirely of amino acids only.

Conjugated or Complex proteins: It is made up of amino acids and other organic or inorganic compounds.

The non-amino acid group is termed as Prosthetic group (e.g.). Lipoproteins—Chylomicrons.

Derived proteins: These are derivatives of proteins resulting from the action of heat, enzymes, or chemical reagents. This group also includes artificially-produced polypeptides e.g. Fibrin.

Classification of proteins (Based on nutritional value)

Proteins are classified into two types based on nutrition viewpoint as follows:

Complete proteins: These contain all the essential amino acids in sufficient quantity to supply the needs of the body. They support life even if supplied as the sole source of protein. These proteins are of animal origin (e.g.) milk, meat, poultry, egg, and fish. The quality of these proteins is much superior to those of incomplete proteins.

PROTEINS

- Simple
  - Globular Proteins
    - Albumins
    - Globulins
    - Glutelins
    - Prolamines
    - Histones
    - Globins
    - Protamines
  - Scleroproteins
    - Collagens
    - Elastins
    - Keratins
- Conjugated
  - Nucleoproteins
  - Glycoproteins Mucoproteins
  - Lipoproteins
  - Phosphoproteins
  - Chromoproteins
  - Metalloproteins
- Derived
  - Primary
    - Coagulated Proteins
    - Proteans
    - Metaproteins
  - Secondary
    - Preoteoses
    - Peptones
    - Polypeptides
    - Peptides

Incomplete proteins: **These proteins are deficient in one or more of the essential amino acids and therefore, they do not support life on their own. All plant sources of proteins (i.e.) vegetables, fruits, cereals, pulses, nuts, and oilseeds contain incomplete proteins to varying degrees.**

COMPLETE VS. INCOMPLETE PROTEINS

Dietary protein is required for the body as there are 9 essential amino acids the body cannot create and must obtain from one’s diet. Complete proteins contain all 9 of these essential amino acids versus incomplete proteins do not.

Complementary proteins are combinations of two or more incomplete proteins that supply all 9 essential amino acids.

Complete Proteins:

→ Meat

→ Poultry

→ Dairy

→ Eggs

→ Fish

Incomplete Proteins

→ Vegetables

→ Grains

→ Legumes/Beans

→ Nuts/Seeds

Complementary proteins: If two sources of incomplete proteins are combined in the same meal, the resulting protein may be of better quality. These are called complementary proteins e.g. Pongal prepared using moong dhal and rice is of better quality than rice or dhal cooked separately. Rice is deficient in the amino acid lysine but rich in methionine. Pulses are rich in lysine but deficient in methionine. So, rice and pulse combination will complement each other. Rice Kheer is another example, where animal and vegetable proteins -milk and rice are cooked together.

Complementary Proteins:

→ Grains+Pulses/Vegetables

→ Nuts/Seeds+Vegetables/Pulses

Food sources of proteins

Milk is a valuable source of protein (casein) although it does not contain a large quantity of protein, the quality is excellent.

Good sources of plant proteins are pulses, nuts, and oilseeds.

CLASSIFICATION OF PROTEIN ACCORDING TO MERITS OF PHYSICAL AND CHEMICAL PROPERTIES

Proteins are classified into three major categories according to their physical and chemical properties, namely

(1) Simple Proteins-Simple proteins are those proteins that are converted into amino acids only after the body's decomposition (Hydrolysis). The following proteins are simple proteins such as:

(i) Albumin: It dissolves in water and coagulates by heat. The lacto albumin of milk is an example,

(ii) Globulin: It is soluble in alkali solution and gets solidified by heat, Such as carcinogens of milk, ovoglobulin of eggs, tuberin of potatoes, etc.

(iv) Prolamin: It is soluble in only 70-80% alcohol. Example: Blydin of wheat, Zein of maize.

(v) Fibrous Protein: These are not soluble in any solution, they are found in the structure of bones, skin, nails, and hair. For example, collagen of tissue, keratin of hair, and myosin of muscle.

(2) Conjugated Protein- Simple protein when combined with another non-protein component, then it is called conjugated protein. Examples of this are the following:

(i) Nucleoprotein—It is found in every cell.

(ii) Glycoprotein-Glycoprotein is found in globulin and albumin of blood.

(iii) Phosphoprotein- is found in milk carcinogen and egg yolk in the form of ovoprotein.

(iv) Haemoglobin-found in the blood by the name of hemoglobin.

(3) Derived Protein-Due to the breakdown of simple and combined proteins in the digestive process, two types of derived proteins are formed before the formation of amino acids. They are primary and secondary derived proteins.

FUNCTIONS OF PROTEIN

Proteins perform the following functions in our body.

1. **The function of body building-**Protein is mainly **a body-building food element**. The presence of protein is followed by the amount of water in the body. For example, protein is mainly present in hair, nails, skin, endocrine glands, muscles, and bones, etc.

Protein is found in all the fluids of the body except bile and urine. It plays an important role in **forming new tissues**, **whereas in infancy, childhood, adolescence, and pregnancy, protein is the main requirement for physical growth and development.**

2. **Work for the reconstruction of the body's wear and tear-**Just as protein helps in the formation of new tissues, in the same way, protein is needed for the repair of the wear and tear caused by the continuous functioning of the body. The wear and tear of cells and tissues in the body can occur during an accident, injury, burn or operation, etc.

3. Blood plasma for a clotting-A protein called fibrinogen is found in the blood plasma for clotting of blood. Bleeding occurs due to injury, cut, or accident. If the blood does not clot within 4-5 minutes and the bleeding does not stop, a lot of blood is lost from the body and life can be in danger. Blood clotting is a complex process. In this process, thrombin, which is a protein converts fibrinogen to fibrin. Due to this, the flow of blood stops. (fibrinogen)

4. Regulating the various functions of the body-Protein keeps balanced between the amount of acid and base in our body regular and when needed, the protein itself acts like a base or acid, which is called Protein Buffers. It does not allow the acid base of the body to accumulate so that the body's pH can remain stable.

5. Regulating the balance of water in the body- Due to the lack of protein, water starts accumulating in the body, because protein maintains the balance of water content on both sides by controlling the activities inside and outside the cell.

6. Producing enzymes, hormones, antibodies- Proteins form many enzymes in the body. It fo many nitrogen-containing compounds in the body, by which many chemical reactions are complete

(a) Manufacture of enzymes—These are made of proteins and proteins form enzymes. Enzy perform various functions in the body; For example, digestion of food, oxidation, metabolism, Enzymes like pepsin, amylase, lipase, etc. are produced only in the presence of proteins. Prot form many nitrogen-rich compounds in the body.

(b) Producing Hormones-Proteins are essential for the manufacture of hormones. Hormones co and regulate various activities. For example, the thyroxine hormone regulates the oxidation process physical growth, insulin plays a main role in maintaining the level of glucose in the blood. Simil the pituitary, adrenaline, gender hormones, etc. perform various functions in the body.

(c) In the formation of antibodies- Antibodies provide the body's ability to fight against various diseases and help in increasing the immunity of the body. Antibodies are not produced due to a lack of protein in the diet. As a result, the ability of a person to fight diseases decreases. A person suffers from many types of diseases and falls ill again and again. Therefore, it is very important to have protein in the food to lead a healthy life.

8. **In providing energy-** in the absence of carbs and fats, protein also serves to provide energy. One gram of protein provides 4 calories of energy. Whenever there is not enough fat and carbs present in the diet, then only protein provides energy. When protein is used in excess, it accumulates in the body in the form of fat and when these are used to provide energy, body-building, which is the main work, becomes secondary.
9. **In milk production-** The mother's milk is the first food of a newborn baby. Mother's milk is the best for the proper nutrition of the baby. Protein is essential for the production of breast milk. Since mother's milk contains 1.2% protein. Therefore, the lactating mother should consume more amount of protein.

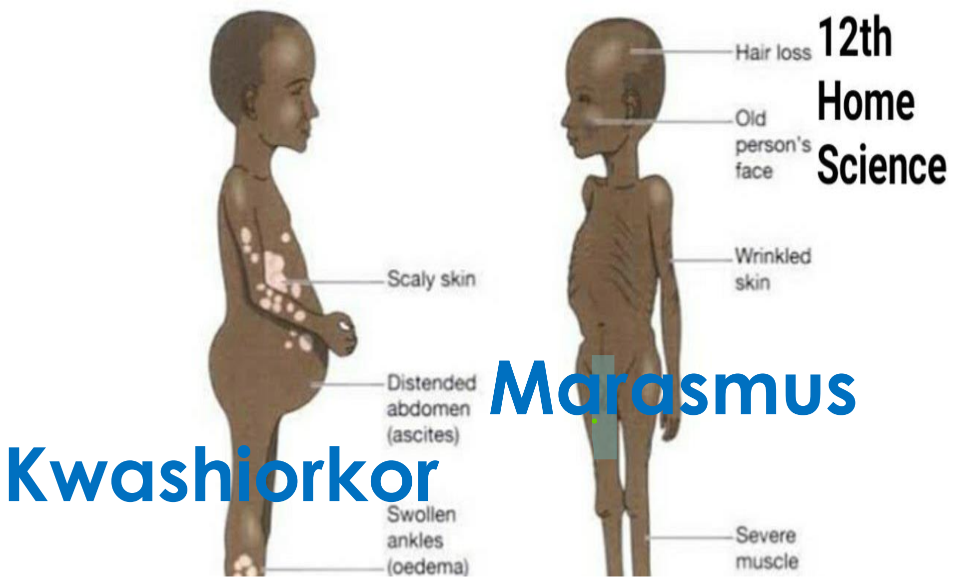
EFFECT OF PROTEIN DEFICIENCY

Protein is very essential for physical growth and development. Protein is the body's constructive element and makes new cells, fibers, and tissues and repairs damaged cells and fibers that are broken. A deficiency of protein leads to anemia, marasmus, kwashiorkor, diarrhea, bloating, etc. The liver does not perform its function properly in the absence of protein.

Most of the victims of protein deficiency are children of 1-5 years. Protein-calorie malnutrition occurs due to a lack of both protein and energy. The following three diseases mainly occur in children due to a deficiency of protein.

1) **Kwashiorkor-**Kwashiorkor was discovered by Sisley William in 1935. It means "the disease that occurs to the first child after the birth of the second child." The following symptoms are seen in children suffering from Kwashiorkor:

2) **Marasmus**—This disease occurs due to a deficiency of both protein and energy. After the age of 6 months, apart from the mother's milk, other food items are also necessary for the children, but if the children do not get them, there is a deficiency in the nutritional elements.



KWASHIORKOR	MARASMUS
✓ Acute illness/infections, measles, AGE, trauma, sepsis are some causes	✓ Severe prolonged starvation, chronic/recurring infections
✓ Protein is principal nutrient	✓ Calories and protein are principal nutrients
✓ 18 months to 3 years	✓ 6 months to 2 years
✓ Rapid, acute onset	✓ Chronic, slow onset
✓ Some weight loss	✓ Severe weight loss
✓ High mortality	✓ Low mortality unless related to underlying disease condition

(3) Wasting of Muscles—The muscles of the body start getting destroyed. Its effect is more on the muscles of the arms, hands, and legs. The arms, hands, and legs become thin and weak.

(4) Mental changes-Apathy and irritability come in the nature of the child. Children start crying while talking and they do not want to play the child feels sluggish, lethargic, and tired.

(5) Anemia-Due to lack of protein, hemoglobin is not formed. Hence. there is a disease of red blood cell deficiency.

(6) Effect of fat in the liver-Due to the lack of protein, fat deposits on the liver, due to which its size increases. The size of the pancreas becomes smaller.

7) Effects on the skin-Due to lack of protein, the skin becomes dry and dull. There is a change in the colour of the skin. There are dark brown and black marks on the skin.

(8) Effect on hair-Hair becomes dry, rough, and dull. There is a change in the colour of the hair. Hair turns gray or white, and hair growth stops.

(9) Effects on Mucous Membrane-Mucous membranes are affected due to lack of protein. Due to this, the lips begin to crack and the surface of the tongue becomes greasy.

(10) Moon-shaped Face—In addition to the hands and feet being swollen, the swelling also occurs on the face, due to which the child's face becomes round like the moon.

(12) Effect on Nervous System-Due to lack of protein, not only the physical development of the child stops, but mental development also stops. The child's ability to learn decreases. Endocrine hormones are not produced and secreted properly. There is a disturbance in blood circulation.

Excess of Protein

1. **Kidney Strain:** Excessive protein intake can put a strain on the kidneys as they are responsible for filtering and eliminating waste products from protein metabolism, including nitrogen. Over time, this increased workload can lead to kidney damage or dysfunction.

2. **Dehydration:** High-protein diets often result in increased water loss through urine. This can lead to dehydration if adequate fluid intake is not maintained.

3. **Mineral Imbalances:** High protein intake can disrupt the balance of minerals in the body, particularly calcium. It may lead to increased calcium excretion through urine, potentially contributing to a **risk of osteoporosis and kidney stones.**

4. **Digestive Issues:** Excess protein can be hard on the digestive system, potentially causing constipation, bloating, and other gastrointestinal discomfort.

5. Increased Risk of Heart Disease: Some high-protein diets, particularly those high in animal protein, may also be high in saturated fats and cholesterol, increasing the risk of heart disease.

6.Weight Gain: Contrary to some beliefs, excessive protein consumption alone may not necessarily lead to weight loss. If the excess protein intake is not balanced with other nutrients and calorie control, it can contribute to weight gain.

7. Bad Breath and Body Odor: Excess protein metabolism can produce ammonia, which can lead to bad breath and unpleasant body odor.

8. Liver Disorders: In rare cases, excessive protein intake can stress the liver and potentially lead to liver disorders.

9.Nutrient Imbalances: Overemphasis on protein intake can sometimes lead to a lack of other essential nutrients, as individuals may prioritize protein-rich foods over a balanced diet containing fruits, vegetables, and whole grains.

DAILY REQUIREMENT OF PROTEIN

The demand for protein is high in infants, children, adolescents, and pregnant and lactating mothers. The demand for protein also increases in the elderly. Indian Council of Medical Research (ICMR) in 1933) nutritionists proposed the daily requirement of protein for Indians as follows:

Daily Requirement of Protein

Stage	*Protein* (grams per day)
1. **Infancy**	
0-6 months	2.05 g/kg by weight
6-12 months	1.65/kg by weight.
2. **Childhood**	
1-3 Years	22
4-6 Years	30
7-9 Years	41
10-12 Years	54
3. **Adolescence**	
13-15 Years (Girls)	65
13 - 15 Years (Boys)	70
16 - 18 Years (Girls)	63
16-18 years (boys)	78
4. **Adult male**	60
5. **Adult female**	
Low active	
Moderate active	50
More active	
6. **Pregnancy**	50 + 15 g
7. **Lactation stage**	
0-6 months	50 + 25 = 75 g
6-12	50 + 18 = 68 grams

Protein content in various food items

Food Ingredients (per 100 g)	*Protein content (in gram)*	*Food Ingredients (per 100 g)*	*Protein content (in gram)*
Animal food		**Dried Nuts**	
Fish	21.8	Groundnut	31.5
Meat	19.8	Cashews	21.2
Liver	19.3	Almonds	18.3
Eggs	13.3	Walnuts	15.6
Milk and Milk Made		Dry Coconut	6.8
		Fresh Coconut	4.2
Buffalo's Milk	4.3	Moong Dal	24.5
Cow's Milk	3.2	Arhar Dal	24.0
Mother's Milk	1.1	Chickpeas	20.0
Curd	3.2	Soyabean	43.2
Mawa	20.1	Cowpea	24.3
Paneer	25.0	Rajma	23.0
Lentil		Black Gram	24
Lentils	25.1	Dry Pea Cereal	19.7
Wheat			
Wheat Flour	12.0	Green Peas	2.0
Maize	11.0	Sweet Potato	1.9
Semolina	10.6	Beet	1.8
Maida	10.1	Potato	1.6
Rice	7.6	Onion	1.2
Vegetables			
Turnip Leaves	2.9	Carrot	1.00
Radish Parima	2.7	Tomato	1.00
Banana	1.00		
Radish	0.7		
Gourd	0.4		

CARBOHYDRATES

Carbohydrates are the combination of carbon, oxygen and hydrogen that can be converted into sugars, chemically these are $Cn(H_2O)n$.

Generally, carbohydrates are divided into three groups according to the sugar group found in them

Monosaccharides (Monosaccharides or Simple Sugars) Mono saccharides are the carbohydrates which are the simplest form of substances that can be digested by the body. They are sweet in flavour and soluble in water. The monosaccharides found in food are called hexoses, which are divided into two groups

Glucose

Fructose

They are converted into simplest sugars by acid, enzymes and heat. These are of three types: (i) sucrose, (ii) maltose, (iii) lactose.

Starch- Plants store carbohydrates in the form of starch, from which they get energy when needed. 65-85% starch is found in cereals, 19.35% in seeds and tubers and 2-10% in fruits and other vegetables. Starch is insoluble in water.

Sources

All types of cereals such as—Wheat, Rice, Millet, Maize, Barley some types of Pulses, Sugar, Jaggery, Honey, Root vegetables, such as potatoes, sweet potatoes, beets, dried fruits; for example, raisins, figs, dates, apples, pears, mangoes, pods, cashews, khoya, cheese etc. are the best means of getting carbs.

Carbohydrate may be:

- **Refined**
- **Unrefined**

Refined means that the food is highly processed. The fiber as well as many of the vitamins and minerals they contain, have been stripped away. Thus, the body processes these carbohydrates quickly and they provide little nutrition although they contain about the same number of calories, refined products are often enriched, meaning vitamins and minerals have been added back to increase their nutritional value. A diet high in simple or

FUNCTIONS OF CARBOHYDRATES

1. **Providing energy/strength**—The main function of carbs is to give energy /strength/heat in the body. Four calories of energy is provided by one gram of carbs. Calories are needed for the proper functioning of the body, brain and nervous system, which is obtained from glucose. Therefore, in both these institutions carbs are involved in providing continuous power. Most of the body activities are related to the activities of the brain and nervous system and depend on carbohydrates for energy.

2. **Making glycogen**-Another function of carbohydrates is to supply the required amount of glycogen in the body. When excess carbohydrate is taken in the body, it converts glycogen into glucose when the amount of glucose is reduced (i) keeps the amount of glucose in the blood normal and (ii) works to give energy/ power in the body. Normally, more than 100 grams of glycogen is stored in the liver of a healthy human. Nerves or muscles store 250 grams of glycogen.

3.Protein saving action-When the body gets less amount of carbs and fat from food items – food items and protein starts working to give power/energy to the body. At this time, protein leaves its function and starts doing the work of carbs and fats, due to which the energy requirement of the body is fulfilled. As soon as the amount of carbs becomes sufficient in the food, the protein becomes free to do its own work. The work of providing strength to the body is considered to be the most important function of the food elements, after which the food elements can do other work.

4. Converting to fat-When the body consumes more amount of carbs than the requirement, all the excess amount gets converted into fat and gets stored in the adipose tissue.

5.Functions of Cellulose-Peristaltic movement comes in the digestive system when cellulose is in sufficient quantity in the diet. Cellulose also helps in water absorption.

6. **Aid in fat metabolism-**Carbohydrates help in complete metabolism of fats and fatty acids in food. Due to the lack of carbohydrates, the metabolism of fats remains incomplete and the amount of acid and ketone bodies in the blood increases, which leads to diseases called acidosis and ketosis.

7. **Functions of Lactose-**Lactose which is found in milk helps in the absorption of calcium in the small intestine. Lactose stays in the stomach for a long time. This leads to the formation of many B vitamins in the stomach. During lactation, the body's glucose is converted into lactose and retains the lactose content of mother's milk.

8. **Maintaining the normal amount of glucose in the blood-**The main function of the carbohydrate substance is to maintain the normal amount of glucose in the blood. The normal amount of glucose in the blood is 80-100 mg glucose per 100 ml of blood. This quantity increases after a meal and when there is a high amount of carbs in the food, a significant increase in the amount of glucose in the blood is seen. It decreases during fasting and disease.

Other functions—The other functions of carbohydrates are:

(i) To manufacture some amino acids

(ii) To provide ribose, a component of nucleic acids

(iii) To make galactosides in the brain tissue

(iv) To digest food and to make flavouring

(v) To provide satisfaction/satiation from food

(vi) To produce vitamins by fermenting bacteria.

EFFECTS OF DEFICIENCY OF CARBOHYDRATE

The amount of carbs in the daily diet should not be reduced. In the absence of carbs and fats, protein starts to provide energy. Due to this, the main function of protein, which is to build muscle, blood, etc., becomes secondary and the work of energy producer becomes main. Due to this the growth and development of the body stops. The following effects are caused by a lack of carbs:

(1) Loss of weight: Deficiency of carbohydrates leads to the loss of protein and fats. Thus it some times reduce the body mass and weight as well.

(2) Feeling tired and weak: Carbohydrate is the primary source of energy for our body. Deficiency of carb make our body tired and weak.

(3) Physical development is stunted: Prolong deficiency of carb restrict body development and growth. Marasmus is a diseases that can be seen in a child with car deficiency along with protein and fat.

(4) Nervousness and irritability in nature: **As carbohydrate is the primary energy source, carb deficiency impacts the nervous system and sometimes makes the child more irritable.**

(5) Decreased body activity: **Body activity depends on the energy obtained from the food. As in the deficiency of carb make body deficient of energy, therefore the activity of the body decreases.**

(7) Wrinkling of the skin

(8) Manifestations of malnutrition

(9) Disturbances in the normal functioning of the nervous system

(10) Disorders related to the digestive system.

EFFECTS OF EXCESS OF CARBOHYDRATE

(1) Excess of carbs in the daily diet leads to obesity. Obesity reduces the flexibility of the body. The working capacity is reduced. Fat people get tired quickly. Obesity leads to diseases like heart disease, diabetes etc.

(2) Consumption of more carbs in food leads to the formation of glucose. Excess glucose gets converted into glycogen and stored in the liver. Excess glucose gets converted into fat, which leads to obesity.

(3) The food is not digested due to the consumption of more carbs in the diet and due to indigestion vomiting, diarrhea also occur.

(4) Eating too much of sugar, jaggery, sweets or chocolate leads to tooth decay.

(5) Consumption of excessive carbs leads to the formation of more glucose. But the capacity of storing glucose in the blood is only 80 mg / 100 ml. The remaining glucose either gets converted into glycogen and goes to the liver, muscles, or it starts coming out with urine, which is a symptom of diabetes.

(6) The pancreas has to work more due to the excess of carbs. A hormone called insulin is released from the the pancreas. This hormone helps in converting glucose into glycogen. The constant release of insulin hormone puts more burden on the pancreas. Therefore, the ability to produce insulin in the desired amount is destroyed, due to which the amount of glucose in the blood remains more than the prescribed amount, which is excreted out of the body through urine. The exit of glucose through urine is called Diabetes or Glucosuria.

DAILY DEMAND OF CARBOHYDRATE

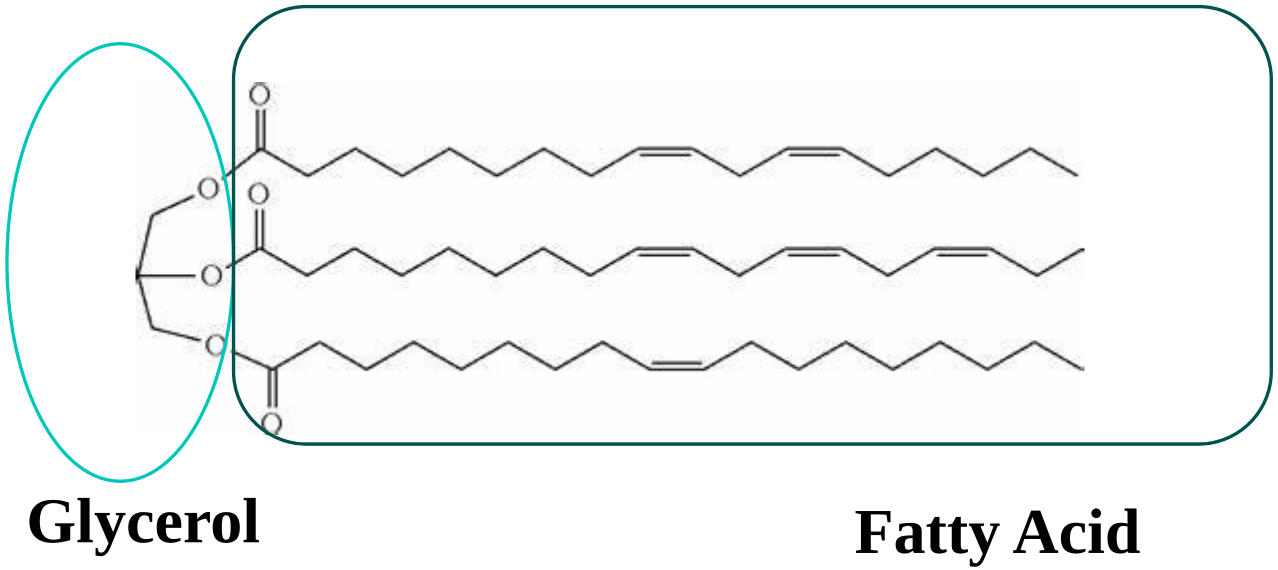
A healthy adult person should take 400–500 grams of carbs in his daily diet. 60-70% of the total body energy demand comes from carbs. Age, gender of the person and the basis of work determines the calorie demand. If the food contains 8 parts of carbs, then at least one part should be protein and one part fat, only then the proper use of protein is done in construction work otherwise protein leaves main work of building the body and starts doing auxiliary work.

People of the rich class get 50-60% of their energy from carbs, while the poor people depend on carbs for energy up to 75-85%. In developed countries America, Japan, Canada like and Russia, people get 40-50% of their energy from carbs and the rest from fat, water (H_2O).

Fats

Fats are oily substances occurring in the animal body and deposited in layers under the skin or around different organs.

Chemically, fats belong to the ester family. Fats are esters formed from the combination of long-chain fatty acids and glycerol.



Fats provide energy of about 9 cal/g. Around 20-30% of total energy is obtained from fats.

Sources of Fats

Table - Fat content in various food ite

Sr. No.	*Name of the food item (per 100 grams)*	*fat (in grams)*	*calories (per 100 g)*
1.	Ghee and oil	100	[illegible]
2.	Vegetable Ghee	100	[illegible]
3.	Fresh Ghee	99.5	895
4.	Butter	81.0	729
5.	Coconut Powder	65.0	600
6.	Coconut Fresh	42.0	370
7.	Almonds	59.3	532
8.	Cashew Nuts	47.0	530
9.	Sesame	43.3	390
[illegible]	[illegible]ndnut	40.1	360
[illegible]	[illegible]ya from buffalo milk	31.2	271
12.	Cow's Milk	36	4.0
13.	Goat's Milk	50	5.5
14.	Mother's Milk	35	3.9
15.	Fish	28	3.2
16.	Meat	119	13.3
17.	Liver	27	3.0
18.	Wheat Embryo	69	7.4
19.	Millet	45	5.0

FUNCTIONS OF FATS

Fat performs the following functions in the body:

1. **Providing energy–**A lot of energy is released from fat. Usually 9 calories of energy is released from one gram of fat.
2. **Storage of energy-**The amount of excess fat in the body is present in the form of adipose tissue below the skin in the form of a layer. At the time of need, this stored fat is used for energy production and when energy is not needed then this fat keeps on accumulating in the fatty fibres, due to which the person becomes fat.
3. **Regulating body temperature-**A normal layer of fat remains under the skin which acts as a bad conductor of heat in the body. Being an insulator, the temperature of the body remains regulated by the environmental influence.
4. **Obtaining fat soluble vitamins-**Fat is necessary for the absorption of vitamins A, D, E, K. If the amount of fat in the diet is low, these vitamins will not be able to accumulate, and various deficiency diseases will occur.

5. Providing essential fatty acids—Some essential fatty acids are not synthesized by the body, it is necessary to supply them through food. The deficiency of essential fatty acids synthesis leads to various skin diseases. It is necessary to take them in good quantity to keep the body healthy.

6.To protect the soft parts of the body-Some essential fatty acids protects soft organs like the kidney, the heart and nervous system from external shocks by forming a deep structure. The highly sensitive organs are also covered with a double layer of fat to protect the soft organs from external shocks.

7. Prolonged satisfaction-Fat slows down the secretion of digestive juices. If the digestive juices are secreted late, then the process of digestion will take time. Therefore, food will remain in the digestive organs for a long time, due to which the person feels the satisfaction of food for a longer time and does not feel hungry.

8. Saving protein-If energy is not supplied by carbohydrates, then protein also does the work of energy production leaving its main building work. In such a situation, if the amount of fat is present in the body, the fat prevents the protein from being used for energy production, thus saving the protein.

9. **To make food tasty-**Fat imparts distinctive taste and smell to food elements. Food is prepared and served in the form of various dishes using fat.

10. **Lubricates the gastric and intestinal tract-**Fat keeps the muscles of the stomach and intestinal tract lubricated (in the form of lubricant) which helps in contraction and release of muscles, conduction of nerve sensations and blood pressure control.

EFFECTS OF DEFICIENCY OF FAT

Due to which the normal growth and development of the body stops:

1. **Normal Growth Stunted-**Deficiency of fats affect activity of amino acids in the body.

2. **Skin Becomes Dry Rough and Lustreless-**Oily glands are present in the dermis layer of the skin. These glands secrete oil continuously, due to which the skin looks smooth and glowing. In the absence of fat, these glands do not secrete sufficient amount of sebum (oil), due to which the skin becomes dry, rough and shine less.

3. **Phrynoderma-**Deficiency of fat causes phrynoderma disease in children and adults. In this disease, sharp pimples appear on the skin of the back, abdomen.

4. Decrease in the Capacity of Cells—The water balance in the body gets disturbed. There is an imbalance in the processes of some enzymes, due to which the speed of cholesterol digestion decreases. Due to this, the amount of cholesterol in the blood increases. Heart diseases occur due to the increase in the amount of cholesterol in the blood.

5. Essential for Reproduction-It has been proved by scientific research that Vitamin ‘E’ is necessary for reproduction. This vitamin is also transported through fat. Due to the lack of fat in the diet, the transport and absorption of vitamin E is not fully done, due to which disturbances in the reproduction process arise.

EFFECTS OF EXCESS OF FAT

The following side effects are seen due to excess of fat:

1. Increase in Obesity-Consumption of more fatty food than necessary leads to the formation of adipose fibers in excess, due to which a layer of fat accumulates under the skin in other soft organs like lungs, heart, testes, liver etc and other places of the body. Due to the accumulation of fat layer, obesity increases. Obesity causes many other diseases like diabetes, high blood pressure, heart disease etc.

Diabetes-Excess intake of fats and carbs leads to the formation of excessive amount of glucose. Only a limited amount of glucose can be stored in the blood. The remaining glucose is converted into glycogen and stored in the liver and muscles. Glucose cannot be converted into glycogen due to the low production of insulin. Due to which glucose starts being expelled from the urinary tract along with urine. This is called diabetes.

Heart Related Disease—The amount of cholesterol in the blood increases due to excess of fat. Due to its accumulation, the amount of cholesterol starts depositing on the inner walls of the arteries. Gradually, cholesterol accumulates on the walls of these arteries. This hardens the walls of the arteries and reduces the flow of blood. This increases the blood pressure, its direct effect falls on the heart and this increases the chances of heart attack.

Effects kidney-Due to obesity, a thick layer of fatty fibers accumulates around the kidney and waste material like urea, uric acid, ammonia, toxins etc. are not completely removed and pigment and harmful substances remain in the blood. As a result, the process of filtering blood is not done properly and pigment and harmful substances remain dissolved in the blood itself. Impure blood is toxic to the body, due to which the person dies. Gradually the kidneys stop working

DAILY DEMAND OF FAT

The body should get a good amount of essential fatty acids through food for he smooth completion of all the functions of fat content in the body. For this purpose, 50% of the total aliphatic substances of food should be taken by vegetable oils.

Generally, the requirement of aliphatic substances of every person is fulfilled by getting a certain percentage of calories from fat substances as follows:

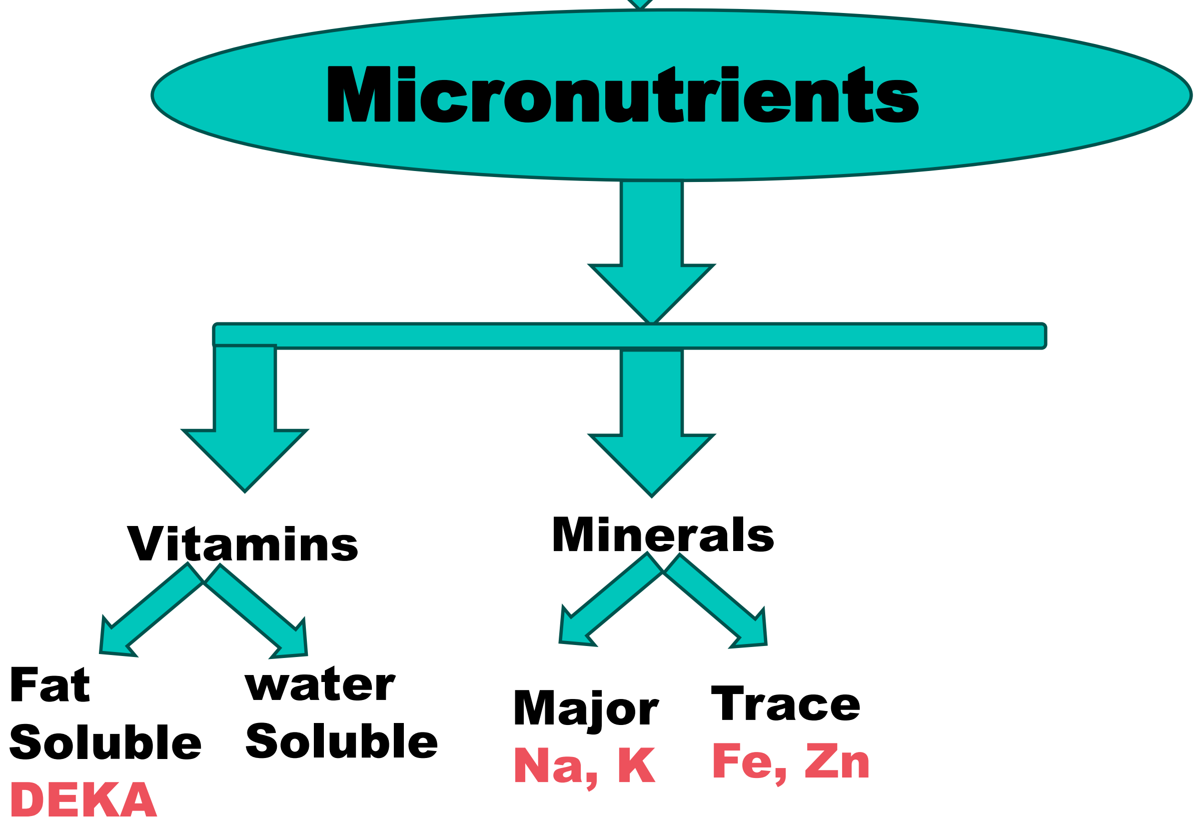
Adult (pregnancy)	10-20% calories from fat
Children 1-18 years old	15-20% calories from fat
Infant (one year old)	25-30% calories from fat
Active adults	30-40% calories from fat

Type of Nutrients

Micronutrients: Micronutrients are essential dietary elements required by organisms in varying comparatively in small quantities throughout life to maintain health.

Humans and other animals require numerous vitamins and dietary minerals. Plants tend not to require vitamins, however minerals are required still.

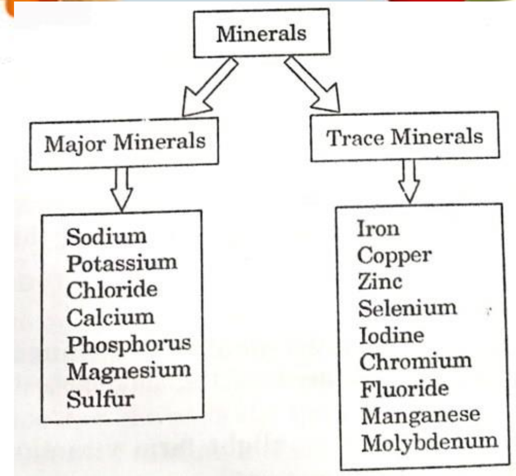
However, these micronutrients are not produced in our bodies and **must be derived from the food we eat**.



Minerals

96% of our total body weight is made up of water, carbs, fats and proteins. The remaining 4% is made up of mineral salts.

Mineral is a naturally occurring inorganic solid with a definite chemical composition and a crystalline structure.



Limestone ($CaCO3$)**, gypsum(**$CaSO_4 \cdot 2H_2O$) **are the most common examples of minerals.**

The growth and development of the body. Of the 4% mineral salts 3/4th part is calcium and phosphorus and the remaining 1/4 part is of other mineral salts.

Properties of minerals

- Minerals are water-soluble and do not require enzymatic digestion.
- Minerals are generally better absorbed from animal-based foods.
- Some minerals influence the absorption of others.
- Minerals are inorganic substances required by the body in small amounts for a variety of different functions.

Functions of Mineral Salts

(1) To structure bone, tooth, tissue, blood and nerve cells.

(2) To regulate the stimulation of contraction and expansion of muscles by nerve fibers.

(3) Maintaining the acid-alkaline balance of fluids/liquids in the body.

(4) To control the water balance.

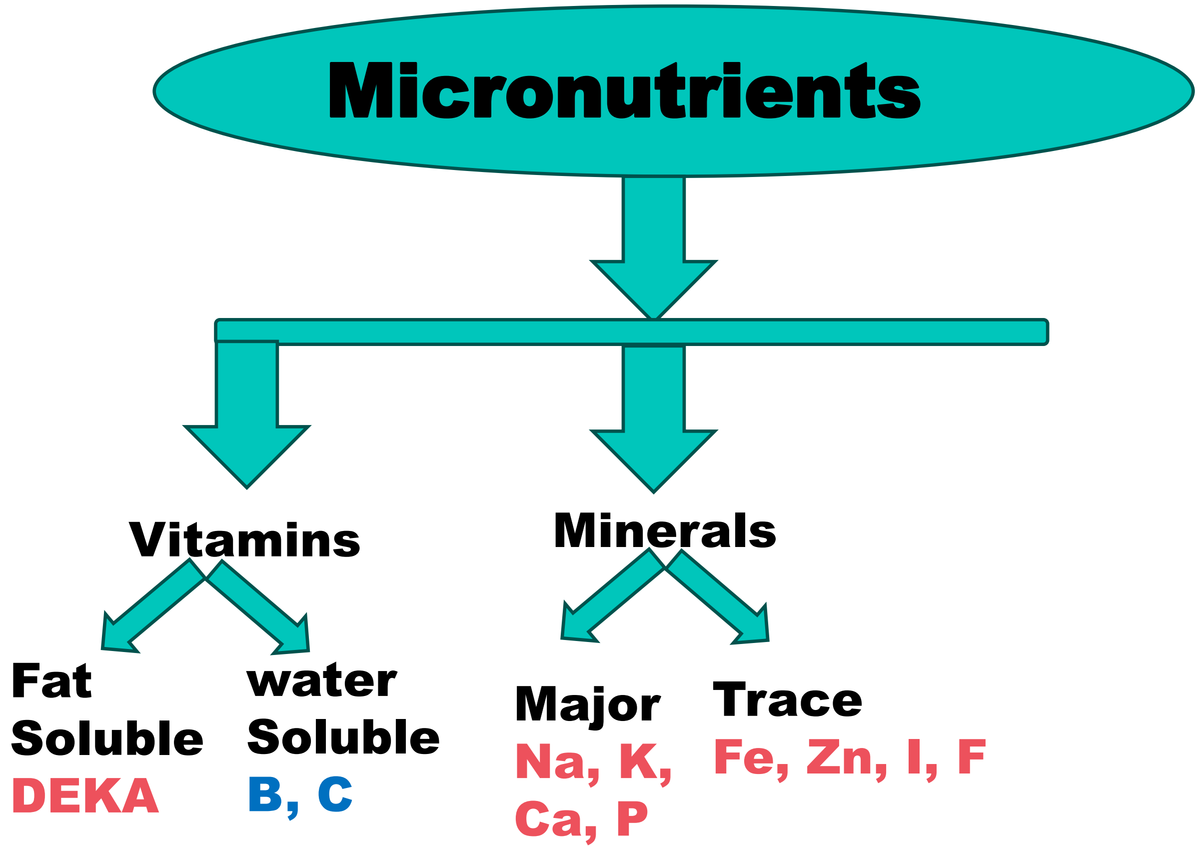
(5) To control the osmotic pressure in the body and the movement power of the cell membranes.

(6) To help in digestion process to increase the usability of food.

(7) Organizing hormones and enzymes.

(8) Activation of enzymes by some minerals.

(9) Controls the oxidation process of cells.


Micronutrients
Vitamins
Minerals
Fat Soluble
DEKA
water Soluble
B, C
Major
Na, K, Ca, P
Trace
Fe, Zn, I, F

Calcium (Ca)

Calcium is the most abundant mineral found in bones and tissues among all mineral salts. About 1200 grams of calcium is found in the body, of which 99% resides in bone, teeth and 1% calcium is found in blood, soft tissues and fluids, whose functions are:

Functions

1.Bone and Teeth Health: Calcium is a major component of bones and teeth, providing structural support and strength. It helps to maintain bone density and prevents conditions like osteoporosis, which is characterized by weak and brittle bones.

2.Muscle Contraction: Calcium ions are essential for muscle contraction. When a muscle is stimulated, calcium is released from storage sites within the muscle cells, allowing the muscle fibers to contract. This process is fundamental for movement, including activities such as walking, running, and lifting.

3.Nerve Function: Calcium is involved in transmitting nerve impulses throughout the nervous system. It helps regulate the release of neurotransmitters (acetylcholine, ions), which are chemical messengers that allow nerve cells to communicate with each other. Proper calcium levels are necessary for the proper functioning of the nervous system.

4. Blood Clotting: Calcium plays a vital role in the blood clotting process, also known as coagulation. When a blood vessel is injured, platelets in the blood release calcium ions, which trigger a series of reactions leading to the formation of a blood clot. This mechanism helps to stop bleeding and promote wound healing.

5. Cell Signaling: Calcium ions act as second messengers in many cellular signaling pathways. They regulate various cellular processes such as cell growth, proliferation, and apoptosis (programmed cell death). Calcium signaling is involved in controlling gene expression, enzyme activity, and hormone secretion.

6. Enzyme Activation: Calcium ions serve as cofactors for several enzymes, meaning they are required for the enzymes to function properly. These enzymes are involved in numerous metabolic reactions within cells, including energy production, DNA synthesis, and protein metabolism.

Effect of Deficiency of Calcium

(1) Deformation of Bones Structure and loss of body weight: **When the amount of calcium that continuously reaches the body is much less than the requirement there is no increase in body weight, there is a hindrance in the development of bones, white spots on the face, white spots on the nails etc, specially in children.**

(2) Impact on nervous system: **When the amount of calcium in the blood decreases to 4-5 mg per 100 ml of serum, the symptoms of tetany disease appear, in which excessive stimulation of the nervous system can lead to unconsciousness and death.**

(3) Ricket Disease: **Rickets has been observed in children due to excessive calcium deficiency. In infants, a lack of calcium and vitamin D leads to a preterm stage due to lack of complete bone growth and development.**

(4) Osteomalacia: During pregnancy, due to the deficiency of calcium and vitamin D, the bones become soft.

Effect of Excess of Calcium

1.Kidney Stones: Consuming excessive calcium can increase the risk of developing kidney stones, particularly if adequate fluids are not consumed to help flush out the excess calcium. Calcium can combine with other substances, such as oxalate or phosphate, forming crystals that can accumulate in the kidneys and lead to the formation of stones.

2.Hypercalcemia: Hypercalcemia refers to elevated levels of calcium in the blood. Excess calcium intake, especially from supplements, can disrupt the body's calcium balance and lead to hypercalcemia. Symptoms of hypercalcemia may include **nausea, vomiting, constipation, abdominal pain, excessive thirst, frequent urination, muscle weakness, fatigue, and confusion**.

3.Interference with Absorption of Other Minerals: Excessive calcium intake, particularly from supplements, can interfere with the absorption of other minerals, such as iron, zinc, magnesium, and phosphorus. This interference can lead to deficiencies of these essential nutrients, potentially causing various health problems.

4. Constipation and Digestive Issues: Excessive calcium intake can cause digestive issues, such as constipation, bloating, and abdominal discomfort, particularly if calcium supplements are taken without sufficient water or food.

5. Cardiovascular Risks: Some research suggests that excessive calcium intake, particularly from supplements, may be associated with an increased risk of cardiovascular events, such as heart attacks and strokes.

Sources of Occurrence of Calcium

There are rich means of obtaining calcium in food items like milk and milk products. Good sources of calcium are green leafy vegetables, peeled sesame seeds, ragi among other Following are the means of obtaining calcium:

Calcium name in food items

Name	*milligram / 100gram*	*Name*	*milligram / 100gram*
Milk	210	Sesame seedless	150
Milk powder	1200	Ragi	350
Paneer	790	Amaranth	500
Khoya	650	Curd	210

Prescribed amount of calcium -1968 as per ICMR

stage	Ca (mg/day)	Stage	Ca (mg/day)
Infant (0-12 months)	500-600	Adults	400-500
Boys (1-9 years)	400-500	Pregnancy	1000
10-15 years	600-700	Menopause	1000
Adolescence (16-19 years)	500-600		

Phosphorus (P)

400-500 grams of phosphorus mineral salts are found in the human body in the form of phosphates. Phosphorus is an important component of RNA and DNA which controls the properties of genes. About 85% of phosphorus in inorganic form works with calcium in bone and tooth structure.

Functions of Phosphorus:

1.Auxiliary in the formation of bones and teeth-Phosphorus combines with calcium to form calcium phosphate (insoluble salts) which make bones and teeth strong. Calcium phosphorous remains in 1:5 ratio in bones. **It is essential for skeletal growth and general health.**

2.Formation of Nucleoproteins and Nucleic acids-Phosphorus is an essential substance for the formation of nucleoproteins and nucleic acids. Nucleoprotein resides in the cytoplasm and nucleus of our cells and here this cell helps in reproduction and exchange of hereditary properties.

3.Maintaining the balance of acid-base in the body—The specialty of phosphorus is that it can combine with more hydrogen iron. Due to this characteristic it prevents the body from being acidic and maintains the balance of acid base in the blood. The amount of phosphorus in the blood serum of adults is 2.5-4 mg / 100 ml and the amount of phosphorus in the serum of children is 4.5 mg / 100 ml.

4. Helps in the energy production process-Phosphorus is an essential component when the liver reuse the glycogen stored in the muscles to get energy, because at that time this glycogen is in the form of a phosphorus-containing glucose compound.

5. Helps in Carbohydrate Metabolism-ATP (Adenosine Triphosphate) is necessary for the metabolism of carbs in cells and for generating heat. Phosphorus combines with other elements to form ATP.

6. Phosphorus—the enzymes involved in the metabolism are needed for the formation of enzymes and co-carboxylase.

7. Facilitates the transport of nutrients-Fats are transported in the blood in the form of phospholipids.

Effects of Deficiency of Phosphorous

1. **Bone Health**: Phosphorus deficiency can lead to weakened bones and increased risk of bone-related conditions such as osteomalacia (softening of bones) and rickets (bone deformities in children). This is because phosphorus is essential for bone mineralization and maintenance of bone density.

2. **Altered Cellular Function**: Phosphorus deficiency may disrupt various cellular processes, including DNA and RNA synthesis, cell signaling, and membrane function. This can have widespread effects on cellular metabolism, gene expression, and overall physiological function.

3. **Impaired Growth and Development**: Phosphorus deficiency can interfere with normal growth and development, particularly in children and adolescents. Insufficient phosphorus intake may hinder bone growth and delay skeletal maturation, leading to stunted growth and developmental delays.

4. **Reduced ATP Production**: Phosphorus is a key component of ATP (adenosine triphosphate), the primary energy currency of cells. A deficiency in phosphorus can impair ATP production, resulting in reduced energy levels and overall fatigue.

5. **Electrolyte Imbalance**: Phosphorus deficiency can disrupt electrolyte balance in the body, leading to abnormalities in fluid balance, nerve function, and muscle contraction. This imbalance may contribute to symptoms such as weakness, muscle cramps, and irregular heart rhythms.

Effect of excess of Phosphorous

1.Calcium Imbalance: High phosphorus levels can interfere with calcium absorption and utilization in the body, leading to a disturbed calcium-phosphorus balance. This imbalance may result in decreased bone mineral density, increased risk of osteoporosis, and bone fractures.

2.Cardiovascular Health: Elevated phosphorus levels have been associated with an increased risk of cardiovascular diseases, including coronary artery calcification, arterial stiffness, and hypertension. Excess phosphorus intake may promote vascular calcification and contribute to the development of cardiovascular complications.

3.Kidney Function: Excessive phosphorus consumption can put strain on the kidneys, as they are responsible for regulating phosphorus levels in the blood. Over time, this increased workload may contribute to kidney damage and impaired renal function, particularly in individuals with pre-existing kidney disease.

4.Mineral Imbalances: High phosphorus intake can disrupt the balance of other minerals in the body, such as magnesium and potassium. Imbalances in these electrolytes can affect nerve function, muscle contraction, and heart rhythm, leading to symptoms like muscle weakness, irregular heartbeat, and fatigue.

5.Bone Health: Although phosphorus is essential for bone mineralization, excessive intake can disrupt the calcium-phosphorus balance and compromise bone health. High phosphorus levels may promote bone resorption and weaken bone structure, increasing the risk of fractures and skeletal abnormalities.

Phosphorus content in various foods

Foods (100gm)	Phosphorus (mg)	Foods (100 gm)	Phosphorous (mg)	Foods (100gm)	Phosphorus (mg)
Buffalo Milk	130	Fish	410	Urad Dal	360
Milk Powder	90	Wheat	320	Peas	300
Milk Powder	730	Maize	330	Coconut (Fresh)	240
Fat Free Milk Powder	1000	Jowar	280	Almond	490
Cheese	520	Bajra	350	Cashew Nuts	450
Khoya	420	Soyabean	690	Groundnut	390
Egg	220	Sesame	570		
Meat	240	Gram Dal	310		

Potassium (K)

The state of potassium mineral salt resides in intracellular fluids, red blood cells and muscles. It is sufficient to take 1.5-6.00 grams of Potassium in equal quantity through daily diet.

Functions of Potassium

(1) Potassium especially works to keep the heart rate regular.
(2) Potassium plays a helpful role in muscle contraction.
(3) It is also helpful in increasing the absorption of some enzymes.

Potassium element is found in milk and milk products. Potassium elements are present in juicy fruits like Lemon, Orange etc. There is less quantity of it in the upper layer of maize flour and rice. Some vegetables and fruits like cucumber, tomato and grapes and potatoes also contain potassium. Potassium is present in all vegetable foods and it is in excess in tea, coffee, cocoa, rice.

Function of Potassium

1. Fluid balance: Potassium helps regulate fluid balance within cells and tissues by maintaining the proper balance of electrolytes, which is critical for normal cell function.

1.Nerve function: Potassium plays a crucial role in nerve impulse transmission. It helps generate electrical impulses that allow nerves to communicate with each other and with muscles. This is essential for proper muscle contraction, including the contraction of the heart muscle.

2.Muscle function: Potassium is involved in muscle contraction and relaxation. It works in tandem with sodium to regulate muscle contractions, including those of the heart.

3.Blood pressure regulation: Potassium helps to regulate blood pressure by counteracting the effects of sodium. Adequate potassium intake is associated with lower blood pressure levels, which reduces the risk of cardiovascular diseases such as hypertension and stroke.

4.Heart function: Potassium is critical for maintaining normal heart rhythm and function. It helps the heart muscle contract rhythmically, ensuring that the heart can effectively pump blood throughout the body.

Excess potassium intake, while less common than potassium deficiency, can still have significant negative impacts on human health:

1. **Cardiac Arrhythmias**: Elevated levels of potassium in the blood, known as hyperkalemia, can disrupt the normal electrical activity of the heart and lead to cardiac arrhythmias. These abnormal heart rhythms may range from palpitations and irregular heartbeat to more severe complications such as ventricular fibrillation or cardiac arrest.
2. **Muscle Weakness and Paralysis**: Excess potassium can affect neuromuscular function, leading to muscle weakness, cramping, and even paralysis. Hyperkalemia interferes with the ability of nerve cells to transmit signals to muscles, resulting in impaired muscle contraction and weakness.
3. **Nausea and Vomiting**: High potassium levels can irritate the gastrointestinal tract, causing symptoms such as nausea, vomiting, abdominal pain, and diarrhea. These symptoms may contribute to dehydration and electrolyte imbalances if left untreated.
4. **Respiratory Distress**: Severe hyperkalemia may affect respiratory muscles, leading to difficulty breathing or respiratory failure. Respiratory distress can be life-threatening and requires immediate medical attention.
5. **Kidney Dysfunction**: The kidneys play a crucial role in regulating potassium levels in the body by excreting excess potassium through urine. However, chronic hyperkalemia can impair kidney function and contribute to the development of kidney disease or exacerbate existing renal conditions.
6. **Cardiovascular Complications**: In addition to cardiac arrhythmias, excess potassium can have other cardiovascular effects, including hypertension and heart failure. Hyperkalemia may disrupt the balance of sodium and potassium in cardiac muscle cells, affecting heart function and increasing the risk of cardiovascular events.
7. **Metabolic Acidosis**: High potassium levels can lead to metabolic acidosis, a condition characterized by an imbalance in the body's acid-base levels. Metabolic acidosis may cause symptoms such as fatigue, confusion, and rapid breathing, and can have systemic effects on various organ systems.

Effect of Potassium deficiency

Muscle Weakness and Fatigue: Potassium is essential for proper muscle function, including muscle contraction. A deficiency in potassium can lead to muscle weakness, fatigue, and cramping, particularly during physical activity. Severe hypokalemia may result in paralysis or muscle breakdown.

1. **Cardiac Arrhythmias**: Potassium plays a crucial role in maintaining the electrical activity of the heart. A deficiency in potassium can disrupt normal heart rhythm and increase the risk of cardiac arrhythmias, palpitations, and irregular heartbeat. Severe hypokalemia can lead to life-threatening arrhythmias such as ventricular tachycardia or fibrillation.
2. **Hypertension**: Potassium helps regulate blood pressure by balancing sodium levels in the body and promoting vasodilation (relaxation of blood vessels). A deficiency in potassium may contribute to high blood pressure (hypertension) and increase the risk of cardiovascular complications such as heart attack and stroke.
3. **Fluid and Electrolyte Imbalance**: Potassium is an essential electrolyte involved in maintaining fluid balance within cells and tissues. A deficiency in potassium can disrupt fluid balance, leading to dehydration, thirst, and electrolyte imbalances. This imbalance can affect various bodily functions, including nerve transmission, muscle contraction, and kidney function.
4. **Weakness and Fatigue**: Potassium deficiency can lead to general weakness, fatigue, and lethargy, as potassium is essential for energy metabolism and cellular function. Individuals with hypokalemia may experience decreased energy levels, difficulty concentrating, and overall feelings of weakness and malaise.
5. **Digestive Issues**: Potassium deficiency can affect gastrointestinal function, leading to symptoms such as constipation, bloating, and abdominal discomfort. Potassium is necessary for proper digestive muscle contractions and may contribute to gastrointestinal motility disorders when deficient.

Sodium (Na)

Its prescribed quantity in the body is indispensable. Combined with chlorine, it forms sodium chloride (NaCl). NaCl in the form of salt is used in the food dishes. Total sodium that remains in the body of a normal person is 92 grams.

Functions of Sodium

(1) The major component of the fluid outside the cell is sodium, which maintains the osmotic pressure and water balance. About 90% sodium is found in the fluid outside the cell.

(2) Sodium works in bringing the contraction of nerves and the activity of nerve cells to normal levels.

(3) Regulates the growth of cells.

(4) Maintains other mineral salts in the fluid inside and outside the cell.

Effect of Deficiency of Sodium

(1) There is a deficiency of sodium in sportsmen and heavy working labor community. Sodium is passed out of the body through sweat. Such a deficiency can be compensated by saline fluids.

(2) Deficiency of Adrenocorticotropic Hormone (ACTH) causes Addison's disease in which excessive amount of sodium is excreted from the body In this condition, the person has a desire to eat more amount of salt.

(3) Due to frequent vomiting and diarrhea, there is a deficiency of sodium in the body,

(4) Sodium deficiency develops into starvation, high blood pressure, heart disease, cirrhosis of the liver and edema. Its fulfillment is done by consuming salt all conditions.

(5) The initial symptoms of sodium deficiency include weakness, dizziness, nausea, lethargy.

Effect of excess of Sodium:

1.Hypertension (High Blood Pressure): One of the most well-known effects of excess sodium consumption is the development of hypertension. High sodium intake can lead to water retention, increased blood volume, and constriction of blood vessels, all of which contribute to elevated blood pressure. Hypertension is a major risk factor for cardiovascular diseases such as heart attack, stroke, and heart failure.

2.Cardiovascular Diseases: Chronic high sodium intake is associated with an increased risk of developing various cardiovascular conditions, including coronary artery disease, atherosclerosis (hardening of the arteries), and cardiac arrhythmias. These conditions can lead to heart attacks, strokes, and other serious cardiovascular events.

3.Kidney Damage: Excessive sodium consumption can strain the kidneys, which are responsible for filtering excess sodium from the bloodstream. Over time, this increased workload can lead to kidney damage and impaired kidney function. Chronic kidney disease is a potential consequence of long-term high sodium intake.

4.Fluid Retention and Edema: High sodium intake can cause the body to retain excess fluid, leading to swelling and edema, particularly in the extremities such as the legs and feet. Fluid retention can exacerbate conditions such as congestive heart failure and liver cirrhosis and contribute to discomfort and reduced mobility.

5.Osteoporosis: Excess sodium intake may increase urinary calcium excretion, leading to calcium depletion in the bones. Over time, this can weaken bone density and increase the risk of osteoporosis, a condition characterized by fragile and brittle bones prone to fractures.

Functions of Iron (Fe)

1.Oxygen Transport: One of the primary functions of iron is to help transport oxygen throughout the body. Iron is a key component of hemoglobin, a protein found in red blood cells that binds to oxygen in the lungs and carries it to tissues and organs throughout the body. Without sufficient iron, the body cannot produce enough hemoglobin, leading to a condition called iron deficiency anemia, which results in fatigue, weakness, and other symptoms due to inadequate oxygen delivery to tissues.

2.DNA Synthesis: Iron plays a role in DNA synthesis, which is essential for cell growth, repair, and replication. Iron-containing enzymes are involved in various steps of DNA synthesis, ensuring that cells can produce new DNA accurately and efficiently.

3.Energy Production: Iron is also involved in energy production within cells. It is a component of enzymes involved in the electron transport chain, which is part of cellular respiration, the process by which cells convert nutrients into energy in the form of adenosine triphosphate (ATP). Iron-containing enzymes facilitate the transfer of electrons during this process, helping to generate ATP for cellular functions.

4.Neurotransmitter Synthesis: Iron is involved in the synthesis of neurotransmitters, such as dopamine, norepinephrine, and serotonin, which are important for mood regulation, cognitive function, and overall brain health. Iron deficiency can affect neurotransmitter synthesis and function, potentially contributing to symptoms such as cognitive impairment and mood disorders.

5.Immune Function: Iron is necessary for proper immune function. It is involved in the production and function of white blood cells, which are crucial for defending the body against infections and pathogens. Iron deficiency can impair immune function, making individuals more susceptible to infections.

Effect of deficiency of Iron

1.Anemia: Iron deficiency is the most common cause of anemia worldwide. Anemia occurs when the body doesn't have enough healthy red blood cells to carry adequate oxygen to tissues and organs. Symptoms of anemia may include fatigue, weakness, pale skin, shortness of breath, dizziness, headache, and cold hands and feet. Severe iron deficiency anemia can impair physical and cognitive function and may lead to complications such as heart problems.

2.Weakness and Fatigue: Iron is essential for energy production within cells. Without enough iron, cells may not produce sufficient energy, leading to feelings of weakness, fatigue, and lethargy.

3.Impaired Cognitive Function: Iron deficiency has been associated with cognitive impairments, including poor memory, difficulty concentrating, and decreased cognitive performance. This is thought to be due to insufficient oxygen delivery to the brain and disruptions in neurotransmitter synthesis caused by iron deficiency.

4.Reduced Exercise Tolerance: Iron deficiency can impair exercise tolerance and physical performance due to decreased oxygen delivery to muscles. This may result in reduced endurance, increased fatigue during exercise, and decreased exercise capacity.

5.Impaired Immune Function: Iron deficiency can weaken the immune system, making individuals more susceptible to infections. This is because iron is necessary for the production and function of white blood cells, which play a crucial role in defending the body against pathogens.

Effect of excess of Iron

1.Iron Overload Disorders: Hemochromatosis is a genetic disorder characterized by excessive absorption and accumulation of iron in the body. In hemochromatosis, the body absorbs more iron than it needs, leading to the gradual buildup of iron in tissues and organs, particularly the liver, heart, pancreas, and joints. This can result in organ damage and dysfunction over time, potentially leading to serious complications such as liver cirrhosis, heart disease, diabetes, and arthritis.

2.Heart Problems: Iron overload can affect the heart, leading to conditions such as cardiomyopathy (disease of the heart muscle), arrhythmias (irregular heartbeats), and heart failure. Excess iron can accumulate in the heart muscle, disrupting its structure and function over time.

3.Liver Damage: Excess iron accumulation in the liver can cause liver damage and dysfunction, leading to conditions such as liver fibrosis, cirrhosis, and eventually liver failure. Chronic iron overload can also increase the risk of liver cancer.

4.Joint Pain and Damage: Iron deposition in the joints can lead to inflammation, pain, and damage, resulting in conditions such as arthritis and joint stiffness.

5.Diabetes: Iron overload has been associated with an increased risk of insulin resistance and type 2 diabetes. Excess iron can impair insulin sensitivity and pancreatic function, contributing to the development of diabetes and metabolic syndrome.

6. Neurological Problems: Excess iron accumulation in the brain has been implicated in neurodegenerative disorders such as Parkinson's disease and Alzheimer's disease. Iron overload can contribute to oxidative stress, inflammation, and neuronal damage in the brain, potentially exacerbating neurodegenerative processes.

Fats soluble vitamins

Vitamin	Sources	Deficiency Symptoms	Effects of Excess Intake
Vitamin K	green leafy vegetables (e.g., kale, spinach, broccoli); fermented foods (e.g., natto, cheese), animal products	Impaired blood clotting, excessive bleeding, easy bruising, bone health issues	Excessive supplementation may interfere with anticoagulant medications, leading to increased bleeding risk
Vitamin D	Sunlight exposure, fatty fish, egg yolks, fortified foods (e.g., milk, cereal)	Rickets, osteomalacia, muscle weakness, immune dysfunction	Vitamin D toxicity: nausea, vomiting, weakness, confusion, elevated blood calcium levels, kidney stones, kidney damage
Vitamin A	Preformed vitamin A: liver, fish, dairy products; Provitamin A carotenoids: carrots, sweet potatoes, spinach, kale	Night blindness, impaired immune function, xerophthalmia, skin disorders	Hypervitaminosis A: nausea, vomiting, headache, dizziness, liver damage, birth defects
Vitamin E	Plant-based oils (e.g., wheat germ oil, sunflower oil, olive oil), nuts, seeds, green leafy vegetables	Muscle weakness, vision problems, neurological abnormalities, reproduction issue, aging issue	Excessive supplementation may interfere with blood clotting and increase bleeding risk